Materials and Engineering Design of the Kronos S.M.A.R.T. Aneutronic Fusion Reactor

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Abstract

Abstract—Kronos S.M.A.R.T. (Superconducting, Minimum-Aspect-Ratio Tokamak) establishes a new paradigm for aneutronic fusion by combining ultra-high magnetic fields, compact spherical geometry, and a **materials-first** design philosophy. Every structural choice is governed by the limits of advanced materials: the brittle ReBCO (Rare-Earth Barium Copper Oxide) high-temperature superconductors degrade above 0.4% strain, so high-modulus carbon fibers (E ≈ 400 GPa), MP35N alloy substrates (σ_y ~1.7 GPa), graphene-reinforced aluminum stabilizers, and refractory ceramic coatings are employed to transform mechanical fragility into robust structural stability[1][2]. Pre-compressed carbon-fiber cocoons, ceramic **bucking cylinders**, and cryogenic "ice" pistons maintain coil integrity under multi-gigapascal Lorentz forces, ensuring the magnets never exceed safe elastic strain. Thermal and radiation challenges are met with multilayer **plasma-facing components**: a HfC/TaC ultra-refractory ceramic armor with 4200 K-class melting point[3], backed by a thermionic emitter (e.g. LaB₆) and actively-cooled structural alloys. These layers directly convert heat to electricity and shield the structural wall from extreme plasma flux. The aneutronic fuel cycle—D–^{3}He for initiation, transitioning to p–^{6}Li → ^{3}He + ^{4}He and ultimately p–^{11}B → 3 ^{4}He—yields >95% of fusion energy in charged particles and X-rays[4]. **Direct energy conversion (DEC)** systems exploit this: electrostatic grids decelerate 3.6 MeV alphas into DC electricity, MHD channels tap plasma exhaust enthalpy, and photon-to-electric devices (thermophotovoltaics and thermionics) capture bremsstrahlung and synchrotron radiation[5][6]. As a result, **net electric efficiency** is engineered to exceed 60% (versus ~35% in conventional steam cycles)[7], dramatically reducing waste heat. The elimination of 14.1 MeV neutron flux (present in D–T fusion) means **minimal structural damage and activation**: first-wall materials avoid the rapid embrittlement and high radio-waste of D–T reactors[8][9]. The **neutron wall loading** is orders of magnitude lower (~0.1 MW/m^2 in D–^{3}He mode vs 2–3 MW/m^2 in D–T)[10], enabling the vacuum vessel and magnets to survive for decades without replacement[11]. High-β plasma shaping (aspect ratio A≈2, elongation κ>2.5, negative triangularity δ<0) maximizes fusion yield per unit volume[12] while maintaining stability through a close-fitting, low-activation conducting wall and active feedback coils. Ignition and sustainment leverage multi-modal heating: 300–500 keV neutral beams, dual-stage Ohmic compression, and RF injection, all managed by AI-assisted plasma control. A **digital twin** of the reactor monitors structural health and optimizes performance via real-time data and physics-informed ML models, predicting material fatigue and guiding preventive maintenance. This integrated approach – weaving together plasma physics, **advanced materials engineering**, and intelligent control – results in a reactor architecture that is compact, efficient, and enduring. The S.M.A.R.T. tokamak is not a mere extrapolation of D–T machines; it is a ground-up reinvention of fusion energy, built around the equations of thermomechanics, plasma-material interactions, and electrical conversion that govern performance and longevity. **Index Terms—**Aneutronic fusion, High-temperature superconductors, Plasma-facing materials, Direct energy conversion, High-entropy alloys, Spherical tokamak, Digital twin, Machine learning.

Index Terms

Aneutronic Fusion; High-Field Tokamak; Plasma-Facing Materials; High-Temperature Superconductors (HTS); ReBCO Tapes; Negative Triangularity; Direct Energy Conversion (DEC); Thermionic Conversion; MHD Generator; High-Entropy Alloys (HEA); Oxide Dispersion Strengthened (ODS) Steels; MAX Phase Ceramics; Yttria-Stabilized Zirconia (YSZ); Hafnia Coatings; Boron Nitride (BN); Aluminum Nitride (AlN); SiC/SiC Composites; Graphene Reinforcement; Digital Twin; Physics-Informed Neural Networks (PINN); Machine Learning (ML); Levelized Cost of Energy (LCOE).

Nomenclature

Symbol / Acronym	Definition
S.M.A.R.T.	*Superconducting Minimum-Aspect-Ratio Tokamak* – Kronos reactor design name.
ReBCO	*Rare-Earth Barium Copper Oxide* – High-*T*_c superconducting tape material (e.g. (Y,Gd)Ba_2Cu_3O₇).
HTS	*High-Temperature Superconductor* – Superconductors (like ReBCO) operating at cryogenic $T > 4$ K (often 20–77 K).
MP35N	A high-strength Ni-Co-Cr-Mo alloy (≈35% Ni & Co) used for coil substrates (E ~230 GPa, σ_y >1.5 GPa).
β (Beta)	Plasma beta – ratio of plasma pressure to magnetic pressure (β = 2μ₀ p / B^2). High β (~30%) indicates efficient pressure confinement.
κ (Kappa)	Plasma elongation – ratio of vertical to horizontal minor radius of plasma cross-section.
δ (Delta)	Plasma triangularity – shaping parameter (δ<0 indicates **negative triangularity** shape).
DPA	Displacements Per Atom – dimensionless measure of radiation damage (average number of times each atom is displaced by neutrons).
appm He	Atom-parts per million of Helium – concentration of helium in materials (from transmutation or implantation).
MTBF	Mean Time Between Failure – average operational time before a component fails or requires replacement.
LCOE	Levelized Cost of Energy – net present cost of electricity generation per unit energy (e.g. $/kWh).
DEC	Direct Energy Conversion – methods to convert kinetic energy of fusion products or photons directly to electricity (bypassing thermal cycles).
MHD	Magnetohydrodynamic – relating to conducting fluids/plasmas interacting with magnetic fields (here, an MHD generator extracts power from plasma flow).
PINN	Physics-Informed Neural Network – neural network that embeds physical laws (differential equations) into its loss function for simulation or prediction.
HEA	High-Entropy Alloy – alloy composed of ~5 or more principal elements in near-equal proportion, offering exceptional strength and radiation tolerance.
ODS Steel	Oxide Dispersion Strengthened steel – ferritic/martensitic steel with fine oxide particles (Y₂O₃ etc.) for improved high-temperature strength and radiation resistance.
MAX Phase	Ternary carbides/nitrides with formula M_{n+1}AX_n (e.g. Ti_3SiC_2, Ti_2AlC) – layered ceramics combining metal and ceramic properties (damage-tolerant, machinable, thermal shock resistant).
YSZ	Yttria-Stabilized Zirconia – ZrO_2 ceramic doped with Y_2O_3, used for thermal barrier coatings and as an electrical insulator in high-temperature environments.
BN (h-BN, c-BN)	Boron Nitride – hexagonal BN (graphite-like, good lubricant and dielectric) or cubic BN (diamond-like, extreme hardness); excellent thermal conductivity and dielectric strength for insulators.
AlN	Aluminum Nitride – ceramic with high thermal conductivity (~>150 W/m·K) and good dielectric properties, used in power electronics substrates and potentially as high-*T* insulating feedthroughs.
SiC/SiC	Silicon Carbide fiber-reinforced composite – a ceramic matrix composite with SiC fibers in SiC matrix, offering high-temperature strength, low activation, and thermal conductivity for structural applications.
Be	Beryllium – low-*Z* metal used in some plasma-facing walls for its low fuel dilution and good neutron moderation (though toxic and brittle).

Symbol / Acronym	Definition
Be_2C	Beryllium Carbide – ultra-refractory compound (melting >2100 °C) potentially forming on Be surfaces or as an advanced coating.
Mo, Rh	Molybdenum, Rhodium – high-Z refractory metals (Mo common in plasma components; Rh exotic, highly reflective and corrosion-resistant, possibly for specialty coatings/diagnostics).
Zerodur	Trade name for a glass-ceramic with ultra-low thermal expansion (used in precision optics); considered for dimensionally stable diagnostic windows or support structures.
GA (GaAs)	Gallium Arsenide – semiconductor used in high-efficiency photovoltaic cells (including those for thermophotonic energy conversion of X-rays).
ARIES	Advanced Reactor Innovation & Evaluation Studies – conceptual fusion power plant studies (e.g. ARIES-III D–^{3}He tokamak).
WDR	Waste Disposal Rating – a measure of how radioactive a material becomes (WDR<0.1 indicates low-level waste suitable for near-surface disposal).

Note: All monetary values are in USD unless stated otherwise. All temperatures in °C (except where Kelvin is used for materials properties). All pressures in Pascals (Pa) or derived units.

I. Introduction

Fusion energy offers the prospect of virtually limitless, clean power. Yet mainstream efforts (ITER, SPARC, etc.) have centered on the **deuterium–tritium (D–T)** reaction, which, while having a relatively low ignition temperature (~10 keV), releases ~80% of its energy in fast neutrons (14.1 MeV)[13][14]. These neutrons pose severe engineering challenges: they displace atoms in structures (causing damage and swelling), transmute materials into radioactive isotopes, and necessitate thick (~1 m) shielding and breeding blankets to absorb their energy and to generate fresh tritium fuel[15][16]. In a D–T tokamak, superconducting magnets must be protected behind heavy shielding to remain cryogenic, making devices large and complex[17]. Even so, critical components like the first wall and divertor suffer cumulative **displacements-per-atom (DPA)** and helium implantation that embrittle metals over time[8][9]. As a result, D–T reactors require frequent replacement of internal components (blankets, first wall, divertors) – often on the scale of every few years – leading to significant downtime and high operations cost[18][19]. The fuel itself (tritium) is radioactive, must be bred on-site, and introduces handling and safety complexities. These factors inflate the **Levelized Cost of Electricity (LCOE)** for D–T fusion plants due to short component lifetimes, extensive safety systems, and downtime penalties.

Aneutronic fusion reactions, in contrast, produce little to no neutron radiation. Reactions such as D–^{3}He (D + ^{3}He → ^{4}He + p, Q = 18.3 MeV), p–^{6}Li (proton + lithium-6 → ^{3}He + ^{4}He, Q = 4.0 MeV), and p–^{11}B (proton + boron-11 → 3 ^{4}He, Q = 8.7 MeV) release energy predominantly in charged particles (protons or alphas) and in electromagnetic radiation[20]. If harnessed, the benefits are transformative: **minimal radioactive waste and structural damage, no tritium breeding requirements, and direct electricity generation via charged particles**[21][22]. The Kronos Fusion Energy S.M.A.R.T. reactor is explicitly designed to exploit these aneutronic fuels, making the machine "neutron-immune by design"[23]. In steady-state operation with advanced fuels (^{3}He–^{3}He or p–^{11}B), the reactor's primary fusion products are charged particles that can be directly converted to electricity, and any residual neutron production is negligible and well-localized[24][25]. The only substantial radiation is in the form of X-rays (from bremsstrahlung and synchrotron emission), which are much easier to absorb or convert than neutrons[26][22]. This *deconflicts* the core reactor design: rather than engineering around neutron damage and activation, Kronos can prioritize **performance** and **efficiency**. Magnets can be placed closer to the plasma without risk of quenching from neutron heating, and the vacuum vessel and first wall can be thin, high-performance structures instead of sacrificial, frequently replaced units[27][28].

The Kronos S.M.A.R.T. design marries this aneutronic **fusion core** with cutting-edge materials science and **direct energy conversion** techniques to reimagine what a fusion power plant looks like. Rather than a massive tokamak blanketed in meters of shielding and driving a steam turbine, S.M.A.R.T. is a compact, high-field spherical tokamak with a multi-layer high-tech wall that *produces electricity at the wall*. Over 95% of the fusion energy emerges as charged particles and photons that are converted *in situ* to electric power[5][29]. The plasma-facing wall is not merely a heat barrier – it is an **energy-harvesting structure** consisting of: (1) a plasma-facing armor (low-Z refractory compound) to handle thermal and particle flux, (2) a thermionic electron emitter converting heat to electricity, (3) photoelectric and photovoltaic layers (GaAs, perovskite cells) to capture x-ray photons, and (4) an actively-cooled heat exchanger for any remaining thermal energy[30][31]. Surrounding the plasma periphery are electrostatic grids and MHD channels that decelerate and divert charged fusion products (like 3.6 MeV alphas from D–^{3}He) to directly generate DC power[32][33]. By intercepting energy *before* it thermalizes, these systems dramatically reduce the heat load on the first wall – for example, capturing ~60% of the power in charged particles means the thermal load is only 40% of the raw fusion power[34][35]. Consequently, a peak wall flux of 40 MW/m^2 would be mitigated to ~16 MW/m^2 of actual heat, a level within the range of advanced materials tests (~20 MW/m^2)[35][36]. This extends component lifetimes and *increases net efficiency* because less coolant pumping and secondary conversion is needed.

Crucially, the S.M.A.R.T. reactor operates in a steady-state or high-duty-cycle mode. Its superconducting magnets (ReBCO tapes) produce an on-axis field of 13–14 T (and >30 T peak at the coil windings) in a compact spherical configuration (major radius R≈2 m)[12]. The plasma has a *quasi-spherical aspect ratio* (A ~ 2.0) to maximize volumetric efficiency, high elongation (κ > 2.5) and **negative triangularity** shaping (δ < 0) to improve stability and confinement[37]. Negative triangularity – a forward-leaning D-shaped plasma cross-section – has been shown in

experiments (e.g. DIII-D) to suppress edge instabilities (ELMs) and allow high performance at high β without large edge transients[38][39]. S.M.A.R.T. is novel in combining spherical tokamak traits (high β, high bootstrap current fraction) with negative triangularity – a synergy made possible by advanced shaping coils. This yields a high-β plasma (target β ≈ 30%) at high field, raising the fusion power density by orders of magnitude compared to conventional aspect ratio tokamaks like ITER[2][40]. In fact, the fundamental **fusion power scaling** roughly follows $P_f \sim β^2 B^4 V$ (since fusion reactivity scales $\sim \propto nT$ and $n \sim \beta B^2$, $T \sim B^2$ for confinement at fixed β, and volume $V \sim R^3$)[41]. By pushing β and B simultaneously to unprecedented levels, S.M.A.R.T. can achieve reactor-level power in a device of ~1/5 to 1/10 the volume of ITER or ARIES-type machines.

The compact, high-field approach does impose **engineering extremes**. The ReBCO coil tapes must endure ~30 T fields and large Lorentz forces while being held below ~0.4% tensile strain (beyond which ReBCO irreversibly loses carrying capacity)[42][43]. Mechanical support is therefore a primary design driver: the coil pack is encapsulated in a stiff carbon-fiber **cocoon** under heavy pre-compression, and a central ceramic **bucking cylinder** carries the inward jacking forces of the TF (Toroidal Field) coils[44][45]. Material selection for these supports emphasizes *ultra-high elastic modulus* and low activation: e.g. Toray M46J carbon fibers with E ~430 GPa[46][47], and hybrid metal/composite structures. High-strength, non-magnetic alloys like MP35N and ODS steels are chosen for bolts, clevises, or reinforcement where some ductility is needed, while ceramics like **silicon carbide composites (SiC/SiC)** or **alumina** provide insulating structural members that withstand both cryogenic temperatures and intense stress. The entire magnet structure is in a deep cryostat (operating ~20 K for coils), which doubles as an radiation shield for any stray neutrons. Notably, because the neutron production is so low, we can place sensors and auxiliary systems closer to the plasma – enabling a rich set of diagnostics and **digital control** loops. High-speed optical cameras (with $MgAl_2O_4$ or YAG transparent ceramic windows), neutron and gamma detectors, and magnetic probes embedded in the wall feed into a machine-learning-based plasma control system[48][49]. **AI algorithms** continuously tune the coil currents, direct converter voltages, and even plasma profiles to optimize performance and avoid instabilities in real time[49].

In summary, the Kronos S.M.A.R.T. reactor represents a bold departure from the mainstream D–T tokamak roadmap. It prioritizes advanced materials and novel subsystems (direct conversion, AI control, high-β shaping) to overcome the plasma physics challenge of aneutronic fuels. The hypothesis is that by *engineering out* the neutron and efficiency penalties of conventional fusion, one can achieve a truly attractive power plant – compact, efficient, with long-lived components and minimal waste. The following sections of this paper provide a deep dive into the **materials science and engineering** of this reactor. We detail how each major system – magnets, first wall, direct conversion, and structural components – is designed around state-of-the-art material capabilities and models. We present the thermal, mechanical, and plasma-interaction models (Fourier conduction, stress/strain analysis, sputtering and sheath physics, etc.) that underpin the design. We introduce the use of **High-Entropy Alloys (HEAs)**, **MAX phase ceramics**, **ODS steels**, and other exotic materials that enable operation in extreme conditions. The integration of machine learning for materials monitoring and a **digital twin** for predictive maintenance is also discussed, highlighting how modern data-driven methods enhance reliability. Finally, we examine the **economic implications**: using a simplified LCOE model, we illustrate how material longevity and high efficiency translate to competitive electricity costs, emphasizing the importance of materials R&D in fusion viability.

A. Key Materials in Kronos S.M.A.R.T. and Their Roles

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
Hafnium Carbide & Hf–Ta Carbides (Ultra-High-Temperature Ceramic)	Melting point 3890–4200 °C[13][14] (highest of known compounds); extremely hard (25–30 GPa) and low sputter yield[15][16]; low thermal expansion and good	Plasma-facing armor on first wall and divertor: tolerates steady ~1700 °C surface temperature[21][22] and transient heat bursts >>10 MW/m^2 without melting[23][24]. Coating on	**Tungsten** (W) armor. Melting 3420 °C but limited to <1200 °C in operation due to recrystallization[21][22]. Hardness ~Vickers 4 GPa (vs 25 GPa for HfC) leading to higher sputter erosion[27][16].

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
	thermal shock resistance[17][18]; moderate thermal conductivity (20–60 W/m·K)[19][20].	structural substrates to protect against erosion and thermal fatigue[25][26].	
Beryllium and Be Compounds (Light Metal/Intermetallic)	Be: low Z, melts ~1285 °C; good neutron moderator; forms BeO surface oxide. Beryllium carbide (Be_2C) is ultra-stable ceramic, high thermal stability. BeO is an excellent insulator with high thermal conductivity.	Not a primary first-wall in Kronos (due to low melting and toxicity), but Be-based ceramics used in select spots: e.g. **BeO liners** as high-temp electrical insulators or radiative coatings near plasma[28]. Be_2C considered for reinforcing Be or as ablative capsule in some fusion designs.	**Beryllium** first wall (ITER). Be offers low plasma dilution but poor high-temp strength. Be surfaces react with carbon (forming Be_2C) and retain tritium; Kronos avoids large Be surfaces and neutron multipliers, focusing on refractory coatings.
Diamond Coatings (Carbon Allotrope)	Exceptional thermal conductivity (up to 2000 W/m·K for single crystal; polycrystalline CVD diamond ~1000 W/m·K), extreme hardness (~100 GPa), low atomic number. Chemically inert; can be made electrically insulating or semiconducting (if doped). Sputtering yield under hydrogen is lower than graphite's by ~50%[29][30].	Thin diamond or diamond-like-carbon (DLC) coatings on plasma probes and mirrors to reduce erosion and outgassing. Diamond-coated diagnostics survive harsh plasma exposure with minimal erosion under detached plasma conditions[31]. Also explored as a high-thermal-conductivity armor or intermediate layer on metals (to spread heat and reduce chemical sputtering).	**Graphite/C tiles** were used in past reactors (e.g. JET) but had high chemical sputtering and tritium retention. Diamond films outperform graphite in erosion resistance[29][30] and can handle >10 MW/m^2 for short exposures[32], making them promising next-gen PFM coatings.

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
CrMoNbV High-Entropy Alloy (Refractory HEA)	Equimolar Cr–Mo–Nb–V forms single-phase BCC alloy[33]. High-temperature strength ~1.06 GPa at 1000 °C (vs <0.1 GPa for 316 steel at 700 °C)[34][35]; retains strength to ~800–900 °C continuous[36]. Excellent creep resistance (orders of magnitude lower creep rate than Fe/Ni alloys at 800–1000 °C)[37][36]. Radiation-tolerant: disordered lattice traps point defects and helium, yielding negligible void	**Vacuum vessel and structural frames:** HEA vessel can operate at 700–800 °C for high-efficiency heat recovery without rapid creep[36][42]. Supports high load at high temperature (ideal for supporting blankets or direct energy converters). Withstands multi-decade neutron exposure with minimal swelling[43][38]. Also used in **bolts/joints** that see ^{3}He bombardment, due to He trapping ability (prevents grain-boundary exfoliation)[39][41].	**RAFM steel (Eurofer)** for vessel/blanket in D–T reactors – service temp limited to ~550 °C[44][45], significant swelling >80 dpa. **Vanadium alloys** (e.g. V-4Cr-4Ti) were studied for advanced blankets: they allow ~700 °C operation and low activation but suffer oxidation and fabrication issues. The CrMoNbV HEA offers a more creep- and radiation-resistant alternative[43][38], with similarly low activation but higher strength.

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
	swelling or He embrittlement up to high fluence[38][39] (nanoscopic He bubbles instead of large voids[40][41]). Low activation elements.		
SiC/SiC Composite (Ceramic Matrix Composite)	Silicon carbide fibers in SiC matrix. Low density, melting >2700 °C, can operate ~1000 °C in oxidizing environment (with protective SiO_2 scale). Electrical insulator (prevents eddy currents). Extremely low neutron activation and very low swelling under neutron flux. Brittle but non-brittle failure modes due to fiber reinforcement; retains strength to very high fluence.	**Insulating structural components:** used as **electrical breaks and thermal insulators** near the plasma and magnets. SiC/SiC liner or tiles can reduce nuclear heating of coils by absorbing radiation while not inducing currents[28]. Considered for **MHD insulator** in direct energy conversion channels (with liquid metal or plasma flow) due to its combination of high-temp stability and electrical insulation.	**Ceramic insulators (Al_2O_3, BeO)** in conventional designs are used for smaller components; **SiC/SiC** is proposed in advanced D–T blankets (e.g. as flow channel insert to insulate liquid metal breeder). Kronos leverages SiC/SiC mainly for its multi-functional use (structural + insulating) in a compact reactor, something not critical in large D–T devices that rely on bulk shield separation.
Vanadium Alloy (V–Cr–Ti)	E.g. V-4Cr-4Ti: Low activation metal with high temperature capability (~750 °C). Good thermal conductivity (~30 W/m·K) and no ferromagnetism. Strength ~400 MPa at 500 °C. Very low swelling under neutron flux <10 dpa, some swelling at higher doses but no phase change. Sensitive to oxygen/nitrogen uptake (embrittlement), requiring high-purity vacuum environment.	**Alternate structural material:** Kronos considers V-alloy for **liquid lithium-facing components** (if a Li breeding blanket or flowing limiter is used) because V alloys have good compatibility with liquid Li and low activation. Could be used for **heat exchanger structures** operating at 700+ °C. Also, vanadium foil is used as a **compliant interlayer** in joining dissimilar materials (e.g. between ceramic and metal) due to mutual solubility with refractory metals and carbon[46].	**None in ITER** (vanadium was not used in ITER/DEMO due to development immaturity). In some advanced reactor studies (ARIES, FNSF), V-4Cr-4Ti was a leading low-activation structural material. Kronos leverages vanadium mainly in supporting roles (e.g. interlayers, or R&D options) rather than as a primary vessel material, given the superior performance of HEAs in most metrics.

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
MP35N Superalloy (Ni–Co–Cr–Mo Alloy)	High-strength cobalt-nickel alloy (approx. 35% Co, 35% Ni, 20% Cr, 10% Mo). Exceptional yield strength >2.1 GPa at room temp, rising to ~2.5 GPa at 77 K[47], with >20% elongation (tough and not brittle)[48][47]. Non-magnetic in annealed condition; corrosion-resistant. Strength and fatigue	**Magnet structural reinforcement:** MP35N is used as **substrate and overband** for HTS (REBCO) tape in the toroidal field coils[49][50]. Its ultra-high strength and good ductility at 4 K allow it to withstand enormous Lorentz forces (>GPa level) while keeping the brittle ReBCO strain <0.4%[49][51]. In Kronos, the	**Austenitic steel** (e.g. 316LN) is often used as a jacket for superconductors in ITER, but yields ~~200 MPa at 4 K – an order of magnitude lower strength[54][55]. **Inconel 718** and similar high-strength alloys (~1 GPa yield) were considered for some magnet parts, but MP35N roughly

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
	performance improve at cryogenic temperatures.	REBCO tapes are mounted on or sandwiched between MP35N strips ("prestressed 30 T magnets" design[52][53]) and wrapped by MP35N or similar high-strength bands. This alloy's toughness ensures **magnet structures can endure cyclic loading** over many pulses without cracking.	doubles the strength and better retains ductility at cryo[48]. The use of MP35N (or comparable Ni–Co alloys) in fusion magnets is novel to Kronos's high-field approach.
Graphene–Aluminum Composite (Metal Matrix Composite)	Aluminum matrix with dispersed graphene nano-platelets (~1 wt%). Graphene dramatically increases thermal and electrical conductivity: e.g. ≥1% graphene doubled an Al alloy's thermal conductivity from ~124 W/m·K to ~280 W/m·K[56][57], and improves electrical conductivity (~66% IACS vs 60% for pure Al)[58][59]. Also raises stiffness and hardness of Al[60]. Retains Al's low density and non-magnetic nature.	**Stabilizer and bus material in coils:** Kronos employs graphene-reinforced Al for **superconductor stabilizer layers and busbars** feeding the magnets[61][59]. By lowering resistive losses (reduced magnetoresistance) and improving heat conduction, this composite helps quench protection – any hot spot can spread heat and current more effectively, preventing damage[61][62]. Also used in **thermal straps** at cryogenic interfaces, where high conductivity and low mass are desired[63][62]. Manufacturing is via powder metallurgy (ball-milling graphene with Al powder, then hot consolidation)[64][65].	**Copper** is the traditional stabilizer in superconducting magnets (e.g. Cu in ITER CS). Al has been used in some high-field magnets (for its low density and higher RRR at cryo), but pure Al has lower conductivity than Cu. Graphene-Al composite matches or exceeds Cu's conductivity at much lighter weight[56]. This is an advanced material unique to modern R&D (TRL ~3–4[66]) that Kronos adopts to push coil performance.
High-Modulus Carbon Fiber Composites (Toray M40J/M46J CFRP)	Carbon fiber reinforced polymer composites with ultra-high Young's modulus (~370–430 GPa)[67] and low thermal expansion (~0, fiber direction)[67]. High strength (~4–5 GPa tensile for fiber, ~1–2 GPa for composite) and excellent fatigue endurance. Fibers are radiation-resistant (carbon self-healing to some extent, though matrix will degrade under neutron flux). Very lightweight.	**Structural supports & coil overwraps:** Kronos uses high-modulus carbon fiber epoxy in **"cocoons" and tie-bandings** that compress and reinforce the magnet coils[68][69]. Wrapped around the TF coil packs, these fiber cocoons apply a pre-compression and take up hoop stress, operating from room temp to cryogenic with minimal thermal contraction mismatch[70][71]. Carbon composite rings and columns also serve in the **central structure** (bucking cylinder) in some designs, either alone or as part of a ceramic composite, to react magnetic forces in compression[72][73]. The near-zero CTE of these composites	**Fiberglass/epoxy** supports (G10, etc.) are used in ITER for magnet tie-rods and clamps, but their modulus is ~20 GPa (vs ~400 GPa for carbon fiber)[75]. Steel keys can provide strength, but add weight and eddy current issues. The use of aerospace-grade carbon fiber in load-bearing fusion structures is a new approach – enabling stiff, non-metallic supports that can handle the ~0.4% strain limits of brittle HTS tapes[76].

Material (Class)	Key Properties	Fusion Role in Kronos	Conventional Analog
		ensures that alignments and preloads are maintained from 300 K down to 4 K without large thermal stresses[67][74].	

Material (Class)	Key Properties	Fusion Role in Kronos	Conventional Analog
Beryllium Carbide (Be_2C) (Ceramic)	Very hard, refractory ceramic (melting ~2100 °C). Formed by reaction of Be with C; has high moderating ratio for neutrons and low absorption. Inert in vacuum; reacts to form BeO in air.	**Diagnostic interfaces & coating:** Investigated for **fiber-reinforced Be composites** (Be_2C fibers in Be matrix) to strengthen plasma-facing Be[77]. In Kronos, Be_2C may appear as a **coating on diagnostic apertures** or mirrors to getter impurities and reduce oxidation. Also considered as a **neutron moderator layer** in certain experimental diagnostics due to its high moderating efficiency (if any stray neutrons need spectrum shifting)[78].	**Beryllides (Be–Ti, Be–V)** – intermetallics studied in fission/fusion for improved high-temp performance over pure Be. Be_2C is analogous in providing higher hardness and stability than Be. Not used in ITER, but research suggests Be_2C coatings can be deposited via plasma processes[79]. Kronos includes it in its materials toolkit for niche roles (acknowledging Be hazards[80]).
Single-Crystal Rhodium & Molybdenum (Metal Mirrors)	Highly reflective metals: Rh and Mo maintain >90% reflectivity over a broad wavelength. Single-crystal form eliminates grain boundaries – improves erosion resistance and limits neutron-induced creep. Rh is noble (oxidation-resistant), Mo has very high melting (2620 °C) and thermal conductivity. Both can be polished to mirror finish and withstand hundreds of °C.	**First mirrors for optical diagnostics:** Kronos employs **single-crystal Rh and Mo mirrors** as plasma-facing optics in diagnostics (for laser interferometry, spectroscopy, imaging). They survive sputtering and impurity deposition better than polycrystalline mirrors – tests show negligible reflectivity loss after 200 plasma cleaning cycles on SC Rh/Mo mirrors[81][82]. Rh is typically used for **visible/IR diagnostics** due to its superior reflectivity and corrosion resistance, while Mo can be used for **UV viewing**. These mirrors may have protective shutters or thin sacrificial coatings (e.g. a removable layer) but the bulk single-crystal material provides long life in vacuum.	**Polycrystalline metal mirrors** (e.g. poly-Mo) were initially considered for ITER, but suffer rapid degradation from sputtering and deposition. ITER's R&D shifted to single-crystal Mo and Rh as the prime mirror materials[83][84]. Kronos follows this state-of-the-art, crucial since aneutronic operation still has plasma erosion (though fewer neutrons to darken mirrors). The difference is Kronos can place diagnostics closer (less neutron shielding needed), so mirror durability is even more critical – hence using the best available mirror materials.
Fused Silica & Synthetic Quartz (SiO_2)	High-purity silica glass (amorphous) and quartz (single crystal). Excellent	**Optical windows and fibers:** Kronos uses **fused silica viewports** on diagnostic ports	**Same in ITER:** Fused silica/quartz is a conventional choice for diagnostic windows

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
	transparency from near-UV to IR. Synthetic grades have minimal impurities and defects, improving radiation resistance (less color-center formation under gamma/neutron irradiation). Fused silica has very low thermal expansion and can handle thermal shocks; quartz has piezoelectric properties.	(to transmit laser beams or observe plasma) due to silica's low thermal expansion and good vacuum compatibility. **Synthetic quartz** is used in **radiation-hardened fiber optic cables and sensor windows** that must maintain clarity over the reactor lifetime. Placing diagnostics close to the plasma (enabled by low neutron flux) means these materials face some radiation and high-temperature bakeouts, for which synthetic SiO_2 is well suited.	(e.g. laser injection for Thomson scattering). The difference in Kronos is potentially higher local heat loads (from direct line-of-sight to plasma X-rays since shielding is thinner). Thus, Kronos may integrate active cooling or choose larger thickness for silica windows. Otherwise, material choice is similar – leveraging the best optical materials from prior fusion projects.
Barium Fluoride (BaF_2) (Optical Crystal)	Wide-bandgap optical crystal with transmission from ~150 nm (UV) to ~11 μm (IR). High density (~4.9 g/cc) and high refractive index (~1.5). Notable for its fast scintillation response to radiation (used in detectors). Moderate hardness (not as durable as quartz) and susceptible to water solvation (needs protection from moisture).	**Radiation detectors & spectral converters:** Kronos includes **BaF_2 scintillator panels** in certain diagnostics to directly convert energetic photons (e.g. fusion gamma/X-rays) into visible light for measurement or even for direct energy capture. For example, a BaF_2 screen paired with photodiodes can monitor the 16.7 MeV gamma from rare D–^{3}He branch reactions. Its fast scintillation (sub-nanosecond decay) suits real-time measurements. Additionally, BaF_2 can serve as a **window for deep UV** spectral diagnostics, as it transmits at wavelengths below fused silica's cutoff.	**Luminoscint detectors** have used BaF_2 in high-radiation experiments, but not in tokamak operation. ITER diagnostics mostly use radiation-hard glass or quartz. Kronos's direct energy conversion architecture, however, inspires use of scintillators like BaF_2 or Ce:YSO to capture otherwise lost photon energy[85][86]. While not a structural material, BaF_2's inclusion highlights Kronos's integrative approach – even optical materials contribute to energy recovery and sensing.
Zerodur (Glass-Ceramic)	Ultra-low thermal expansion glass-ceramic (α ~ $0 \pm 0.02\times10^{-6}$ K^{-1} near 20 °C). Highly stable dimensionally under temperature changes. Good thermal stability and can be polished to optical flatness. Reasonably radiation resistant (developed for telescope mirrors, not extensively tested in neutron fields but should see minimal changes at low fluence).	**Metrology and support optics:** Kronos uses **Zerodur components in laser diagnostic systems** that require extreme dimensional stability – for example, reference mirror mounts or optical bench structures near the reactor that cannot deform with temperature. It might also serve as the substrate for large plasma-facing mirror assemblies (kept just outside the vacuum) to ensure	**N/A in tokamaks** – Zerodur is known in astronomy but not previously needed in reactor diagnostics. Kronos's use case comes from the need to maintain precise alignment for laser diagnostics in a compact reactor that may see rapid thermal ramps. By anchoring sensitive optics to Zerodur bases, Kronos ensures calibration stability. This deviates from simpler supports (steel or aluminum) in conventional systems,

Material (Class)	**Key Properties**	**Fusion Role in Kronos**	**Conventional Analog**
		alignment is maintained during reactor pulse thermal cycles.	reflecting the high precision required in Kronos's integrated design.
316L Stainless Steel (Austenitic Steel)	Fe–Cr–Ni–Mo alloy; non-magnetic, good general corrosion resistance. Yield strength ~200 MPa at RT, drops at high temperature (<100 MPa at 700 °C)[87][55]. Under neutron irradiation, 316L experiences significant activation (due to Ni, Mo) and swelling at high dpa, but low-carbon "L" grade minimizes irradiation creep. Well-characterized and weldable.	**Ancillary structures and vacuum vessel (outer layers):** While advanced alloys replace 316L in most high-stress internal roles, Kronos still uses 316L for **secondary structures** – e.g. outer vacuum chamber or cryostat wall, coolant piping, diagnostic housings – where its ease of fabrication and existing ASME code qualification are advantageous. It's also used in **instrumentation feedthroughs** and flanges due to its reliability in ultra-high vacuum and compatibility with industrial vacuum hardware. Kronos's aneutronic operation means the activation of 316L (e.g. Co -> Co-60) is much reduced, so small amounts can be tolerated[88][89].	**316L(N) in ITER**: ITER uses 316LN for the vacuum vessel shell. Kronos similarly might have a steel outer shell, but notably, **internal structures are not 316** (unlike some experimental tokamaks) – they are replaced by HEA and composites. The minimal neutron output of aneutronic fusion makes the penalty of using a few conventional steel components negligible[88]. However, Kronos's core design shuns 316L in high-heat regions due to its low high-T strength, which is a departure from many existing test reactors where SS316 structures are common.

Figure 1. Cross-sectional schematic of the Kronos S.M.A.R.T. tokamak highlighting material deployment (not to scale). The plasma-facing first wall is coated with an ultrarefractory hafnium carbonitride ceramic, backed by a high-entropy alloy structural wall panel. The divertor region (bottom) uses advanced tungsten-based composite armor with thermionic converters. The toroidal field (TF) coils use REBCO superconductors supported by MP35N metal backing strips and wrapped in high-modulus carbon fiber "cocoons"[68][69]; a central ceramic **bucking cylinder** made of a high-strength composite resists the inward magnetic pressure[90][91]. Diagnostic ports around the midplane incorporate single-crystal molybdenum mirrors and fused silica windows, mounted on Zerodur frames for stability. Notably, a thin liquid lithium blanket (optional in some designs) lines the interior behind the first wall to breed ^{3}He (via p–^{6}Li reactions)[92][93], which is extracted for fueling – this liquid is contained by a SiC composite channel to provide electrical insulation for MHD direct conversion.

II. Reactor Architecture and Materials-First Design

A. Compact Spherical Tokamak Configuration

S.M.A.R.T.'s geometry is a low-aspect-ratio (**spherical tokamak**) with major radius R = 2.0 m and minor radius a ~ 1.0 m (aspect ratio A ≈ 2). This quasi-spherical shape offers a very high **plasma beta** (pressure fraction) for a given magnetic field. Experiments like NSTX (A ~1.3) achieved β ~ 40%[50], vastly higher than conventional aspect ratio tokamaks (which typically max out at β ~5–10%). Operating at high β allows S.M.A.R.T. to produce more fusion power for a given magnet size – essential for aneutronic fuels which require high plasma temperatures and densities. However, high β can induce instabilities (kink modes, ballooning modes) that must be controlled. The design addresses this with a close-fitting conducting wall (the vacuum vessel) and active feedback coils located just behind the first wall[51][52]. The conducting wall, being only a small fraction of the plasma radius away, provides passive stabilization (it raises the threshold for wall-stabilized kink modes). In fact, because S.M.A.R.T.'s magnets are not threatened by neutrons, we can afford a very minimal gap between plasma and wall – improving stability compared to ITER/ARC where a thick blanket separates plasma from any conducting structure[51][27]. Additionally, negative triangularity (δ < 0) shaping has been experimentally observed (on DIII-D, TCV, etc.) to suppress edge-localized modes (ELMs) and improve energy confinement even at high β[53][54]. S.M.A.R.T. uses δ ~ -0.3 to -0.5 (plasma bottom wider than top), which also increases room in the divertor region for direct converter hardware[55][56]. If needed, the configuration is flexible: shaping coils can adjust δ back toward 0 or positive values if negative triangularity poses unforeseen problems at spherical aspect ratio[57][58].

Magnetic field strength is S.M.A.R.T.'s other key advantage. The central TF magnet reaches >30 T at the coil (using ReBCO tapes), giving ~13–14 T on-axis field in the plasma[37][59]. For comparison, ITER's on-axis field is 5.3 T and ARC's is ~9 T; 13 T is a record high for any reactor-scale device. Fusion power density scales roughly as ~ B^4 in the empirical range, so this boost in B multiplies the power output tremendously (all else equal). High field also allows achieving the ~100 keV ion temperatures needed for p–^{11}B or D–^{3}He at reasonable plasma pressures and confinement times. However, **engineering the TF coils at 30 T** is non-trivial. The Lorentz *hoop stress* in a tokamak coil can be estimated by treating each toroidal field coil as a hoop under magnetic pressure $p_B = B^2/(2μ_0)$. At B = 30 T, $p_B ≈ 360$ MPa (about 3.6 kN/cm^2). For a coil of radius ~2 m, the total inward force per meter of height is $F = p_B R ≈ 720$ MN/m. These forces tend to make the TF coils expand outward and also crush inward on the center stack. Additionally, there are strong **out-of-plane forces** (during plasma disruptions or due to the shape of the coils). S.M.A.R.T. addresses this with an integrated mechanical structure:

- **Reinforced HTS Tapes:** Each ReBCO tape in the winding pack is a multi-layer composite engineered for strength. As shown in **Figure 1**, the tape uses an MP35N alloy substrate (230 GPa Young's modulus) instead of a conventional Hastelloy (which is ~200 GPa)[60][61]. MP35N provides not only high stiffness but also high yield strength (>1.5 GPa)[62]. A thin textured oxide buffer (e.g. IBAD YSZ) isolates the superconductor layer but transmits strain; atop that sits the ~1.5 μm ReBCO film (which can carry critical current densities J_c ~20 MA/cm^2 at 20 K, 20 T[63][64]). The ReBCO is capped with a silver layer (~2 μm) that serves as a solderable contact and a ductile strain relief[65][66]. Finally, a thick copper stabilizer (~50–100 μm) and embedded high-purity aluminum strips run along the tape[67][68]. The aluminum (often graphene-reinforced Al) acts as a **bypass conductor** – if any segment of ReBCO normalizes (quench), current can rapidly redistribute into the Al and Cu, preventing local overheating[69][70]. A unique innovation is a thin coating of vanadium oxide (V₂O₃) on the tape, which is insulating at low temperature but becomes conductive if heated, thus providing an automatic quench current path that kicks in *only* when needed[71][72]. In short, each HTS tape is a high-performance composite designed to **limit strain** on the ReBCO layer to <0.4% under full load[73] and to **mitigate hotspots** via current sharing[74][56]. By using these tapes, S.M.A.R.T. can sustain 30 T fields without the superconductor failing – a result confirmed by recent 20 T large-bore magnet demos (e.g. MIT PSC/CFS 2021)[75][76].

- **Carbon Fiber Support Structure:** Surrounding the coil winding pack, all key structural supports are made of **ultra high modulus carbon fiber-reinforced polymers (CFRP)**. Toray M40J and M46J fibers (modulus 377–430 GPa) are used in epoxy or cyanate ester matrices to form load rings, tie rods, and an external

"cocoon" shell[77][78]. The CFRP components have tensile strengths in excess of 4 GPa[77]. They are essentially "wrapping" the coil to prevent it from expanding. The external cocoon is a continuous filament-wound shell that is applied under tension (pre-loading the coils in compression)[45][79]. Think of it like a **pre-stressed concrete** analogy: the carbon shell squeezes the coil so that when the magnetic force tries to push it outward, it first just relieves some pre-compression but does not put the superconductor in net tension[80][81]. This keeps actual strain in the ReBCO near zero throughout the pulse. The cocoon also provides fatigue benefits: it prevents cyclic micro-motion or "rattle" of coil segments, reducing accumulation of strain damage over many pulses[81][82]. Carbon fiber is chosen not only for its modulus but also for its radiation transparency and low activation. In a neutron-rich reactor, carbon would pose issues (it can form C-14), but in S.M.A.R.T. the neutron fluence is so low that this is negligible (and carbon's activation products are short-lived anyway[83]). Moreover, carbon composites maintain strength at cryogenic temperatures and are extremely lightweight. The **low density** helps because it minimizes the mechanical inertia and seismic loads on the magnet system.

- **Central Bucking Cylinder:** Inside the doughnut of the tokamak, along the central column, is a thick-walled **bucking cylinder** that reacts the inward forces. We choose a high-compression-strength, nonconductive material – candidate options include *silicon carbide composite (3-D woven SiC/SiC)* or *sintered alumina toughened with zirconia (ZTA)*. This cylinder runs inside the inner legs of the TF coils (or could form the inner vacuum vessel wall) and carries the compressive load that each pair of TF coils exert towards each other[44][84]. By providing this "backbone," the TF coils do not have to support all the inward pinch themselves, avoiding excessive interturn compression that could crack insulation or the ReBCO tapes[85]. Finite element analysis indicates that a properly designed bucking post can cut peak coil stress by a large factor[85]. SiC/SiC is attractive here because it has >500 MPa compressive strength even at high temperature, negligible activation (Si and C produce only short-lived isotopes), and it is an electrical insulator (so it won't induce large eddy currents during fast transients). Its low thermal expansion also helps maintain alignment through cooldown and operation. Alternatively, a glass-ceramic like **Zerodur** or a fiber-wound ceramic (e.g. ZrO_2 fiber composite) could be used if we need more elastic yield. The bucking cylinder must also handle vertical forces (transmitted from the coils via shear keys or spokes), making toughness and fracture resistance important – hence a composite rather than monolithic ceramic is preferred.

- **Ice-Expanded Support Pistons:** A novel feature in Kronos is the use of cryogenic **ice pistons** to preload the structure against vertical forces[86][87]. The D-shaped TF coils experience an "out-of-plane" force tending to elongate them (imagine the top and bottom of the D want to spring outward when current flows, trying to straighten the D). This concentrates stress at the inner top/bottom of the coils (the "throat"). To counteract that, we place adjustable jacks between the coil and support structure at top and bottom. These jacks are filled with a fluid (water or a gas) at assembly at room temperature, then sealed. When the reactor is cooled to cryogenic temp, the fluid freezes and expands ~~9% (in the case of water turning to ice)[88][87]. This expansion pushes against the coil supports, creating an **upward preload** that opposes the Lorentz deflection. Essentially, by the time the magnet is energized, the structure is already braced – the coils "push" on a support that's already pushing back, limiting net displacement[89][90]. The beauty of this approach is that it's *passive and self-regulating*: the amount of expansion (hence preload) is set by the phase change volume difference, which is consistent and temperature-triggered. There are no moving parts once frozen, and if the coil warms (losing current), the ice may partially melt/contract, preventing overstrain. This technique is akin to using thermal shrink-fit or smart materials to achieve an optimal stress bias. It leverages cryogenic cooling (a "free" resource in a superconducting magnet system) to actuate mechanical reinforcement.

- **Helical Winding Pitch:** The TF coils are given a slight **helical twist** (~few degrees over their height) rather than being perfectly planar rings[91]. This means each turn is not exactly in the poloidal plane but slants, distributing forces differently. The helical pitch causes some fraction of what would be pure vertical force to be borne as additional toroidal tension or vice versa, effectively spreading out loads and avoiding singular concentration at one point[91][92]. It also couples the TF and PF (poloidal field) roles slightly – that is, the TF

coils with a twist generate a small poloidal field component, which can help share shaping loads. This "blurred" distinction allows the TF coils to assist in equilibrium, reducing demands on separate shaping coils[92][93]. Moreover, from a materials perspective, a multi-axial stress distribution might be easier to handle than a single-axis extreme. The composite structure can be optimized for a combination of tension and bending, rather than pure bending at one location. We ensure that the helical tweak is small enough not to complicate coil fabrication or plasma symmetry significantly.

Overall, these design features illustrate the **materials-first philosophy**: rather than selecting off-the-shelf components, Kronos engineers each aspect of the magnet and structure to use the best available materials in an optimal way. *High-modulus, high-strength materials (MP35N, M46J carbon fiber)* carry the brunt of forces to keep superconductor strain low[43][47]. *High-conductivity materials (Cu, graphene-Al)* are integrated for quench safety. *Smart coatings (V_2O_3)* and *thermal actuators (ice pistons)* provide passive protection and load management. And critically, all materials were chosen with an eye on **nuclear compatibility**: low neutron activation and tolerance to any radiation present. For example, MP35N alloy's constituents (Co, Ni, Cr, Mo) are not all low-activation by fission standards, but in an aneutronic reactor the total fluence is so low that we expect it to qualify as Class A low-level waste at end-of-life[94][95]. Many of the structural components (fibers, ceramics) are composed of elements (C, Si, O, Al) that produce only short-lived isotopes (like C-14, which has half-life ~5k years, but in small quantities)[9][83]. If S.M.A.R.T. eventually transitions to pure p–^{11}B fuel, neutron production would drop essentially to zero (aside from trace side reactions), meaning the magnet structure could potentially **last the entire plant lifetime** (several decades) without needing replacement[11][96]. This is a stark contrast to D–T machines where magnets might be lifetime components but are kept far from the plasma; here magnets are near plasma but fine because the plasma is aneutronic.

Finally, **plasma control** benefits from the compact, high-field setup. The small size yields fast current ramp-up and plasma dynamic times (<1 s) which, coupled with full diagnostics coverage, make it feasible to implement advanced feedback schemes. Kronos plans to incorporate machine learning for controlling plasma instabilities and optimizing performance on the fly[48][49]. For instance, an ML-based disruption predictor will monitor hundreds of signals (densities, mode amplitudes, etc.) and adjust actuators (like ECCD or vertical stability coils) to avoid disruptions. In high β operation, resistive wall modes could arise; active mode control coils under the blanket, driven by real-time AI controllers, can apply correcting fields in <1 ms. The close wall and high field also raise the natural **plasma rotation and alpha stabilization** – fast alpha particles in a strong B field can stabilize sawteeth or resistive instabilities via kinetic effects, a bonus of aneutronic alphas versus Maxwellian neutrons of D–T. Overall, the architecture combines passive stability from geometry/materials with active stability from intelligent controls.

B. Plasma-Facing Components and First Wall Design

The **first wall (FW)** and **divertor** in any fusion reactor face extreme heat and particle flux. In S.M.A.R.T., because direct conversion intercepts much of the charged particle energy, the heat loads are moderated, but they are still significant. We anticipate steady-state thermal fluxes of order 5–10 MW/m^2 on the divertor strike points and perhaps 1–5 MW/m^2 on the main chamber first wall[97][98]. Transient events (if any ELM-like bursts or ripple losses occur) could locally spike higher, though the negative triangularity regime is expected to suppress large ELMs. The design uses a **multi-layer plasma-facing component (PFC)** strategy, as illustrated conceptually in **Figure 2**.

- **Plasma-Facing Armor:** The top layer is a refractory, **low-Z ceramic** chosen for its high temperature tolerance, sputtering resistance, and low fuel contamination. We eschew tungsten (Z=74) as the primary armor because even though W has a high melting point (3420 °C), tungsten is high-Z and a heavy impurity in a plasma can cause excessive radiative losses if eroded (and in aneutronic plasmas with very low intrinsic radiation, a heavy impurity could quickly quench the plasma). Instead, Kronos uses **hafnium carbide (HfC)** and **tantalum carbide (TaC)** composites as armor tiles[99][100]. These are ultra-high-temperature ceramics with the highest melting points known (HfC≈4232 K and TaC≈4150 K)[3]. HfC is a bit paradoxical since Hf is high-Z (Z=72), but in carbide form it is extremely refractory and has a *very low sputtering yield* for hydrogen/helium ions – so it erodes much more slowly than graphite or Be for a given flux[101][102]. The key is that HfC's sputtered Hf atoms are heavy and have difficulty entering the core plasma (they tend to localize

and promptly redeposit). Additionally, any Hf that does enter is highly radiating but because the plasma is already designed for high power density, a small fraction won't be disastrous – and we expect to keep the net impurity level below ~1e-5. TaC is similar and often mixed with HfC to form a solid solution (incidentally, mixing can raise the sublimation point further)[103]. These carbides can handle continuous surface temperatures ~>3000 °C, so even at 10–20 MW/m^2, with active cooling they should survive (a 20 MW/m^2 heat flux into HfC might result in ~2000 °C surface if cooled on backside at 600 °C, manageable for HfC). They also exhibit good thermal conductivity among ceramics (~20 W/m·K at high temp), better than pure ZrC or SiC, which helps spread heat. A potential issue is thermal shock and grain cracking – to mitigate that, small tiles are used (~5 cm scale) to reduce thermal stresses, and the HfC is possibly graded or composite (e.g. HfC matrix with fibers, or functionally graded to a metal substrate) to reduce CTE mismatch.

- **Thermionic Electron Converter:** Immediately behind the armor, a **thermionic emission layer** converts heat to electricity. This layer can be, for instance, a **lanthanum hexaboride (LaB₆)** emitter coating on a refractory metal substrate (like molybdenum or tungsten alloy base)[104][105]. LaB₆ has a low work function (~2.7 eV) and can emit copious electrons at temperatures above ~1500 °C. The concept is that as the armor heats to 1500–2000 °C under plasma flux, it drives a thermionic current from this emitter into a collector grid (biased at a few volts positive relative to the emitter). Each tile effectively becomes a tiny cathode of a vacuum diode, outputting DC current. Thermionic conversion can directly yield 10–20% efficiency in converting heat to electricity at these temperatures[106][107]. By incorporating it, we reduce the heat flowing into the coolant and simultaneously generate power. The refractory substrate for LaB₆ could be something like **TZM alloy** (Mo-0.5Ti-0.1Zr) which is stable up to 1400 °C or **ODS Tungsten** if tungsten is acceptable in small amounts. However, tungsten as a substrate is somewhat acceptable since it's not directly in contact with plasma (the HfC is) and any W ions from substrate would have to diffuse through cracks or edges. We prefer molybdenum or **alloys like Mo-30%W** which have slightly lower Z and good high-temp strength. *Why LaB₆?* It's a proven cathode material (used in electron microscopes, etc.), can sustain >2000 °C, and B and La have intermediate Z (La is 57 but locked in compound with low sputter yield, B is 5). Alternatives like Y₂O₃-doped Ir or CeB₆ could also be considered. The collector for the thermionic layer might be an interwoven metallic grid slightly above the surface or perhaps the coolant tube itself if biased correctly.

- **Photonic Energy Conversion (TPV/PV):** The plasma and hot walls emit **bremsstrahlung and line radiation** (X-rays, UV, visible). Instead of letting all this thermal radiation become heat in the blanket, Kronos employs **thermophotovoltaic (TPV) cells** or specialized **photoelectric converters** behind the thermionic layer[5]. For example, a thin **GaAs solar cell** optimized for X-ray/UV frequencies or a multi-junction cell with a perovskite top layer can directly absorb photons from the ~2000 °C walls (peak emission ~1 μm wavelength, plus shorter-wavelength X-rays from the plasma). These cells could be mounted on a heat sink at say 200 °C, converting a portion of the radiation to electricity with efficiency perhaps 20–30% of the photon power. Additionally, **photoelectric panels** (like a tungsten or CsI coated plate under bias) can absorb X-rays and eject electrons, similarly to a solar panel concept for X-rays[23][22]. The Kronos whitepaper alludes to "photon-absorbing GaAs/perovskite photovoltaics" built into the wall[5]. This is certainly a bleeding-edge concept (no current reactor uses PV in the blanket), but research in TPV (for capturing waste heat from furnaces) has made progress with cells that can survive high radiation flux. Another approach is using **fluorescent converters**: X-rays hit a layer of material (like a scintillator or phosphor) that re-emits visible light, which is then converted by standard PV. For instance, a thin layer of europium-doped YAG or a MAX phase like *Y₃Al₅O₁₂* could serve as a robust scintillating layer (YAG is also a transparent ceramic that can handle radiation). The optical part of the wall needs to be transparent to the relevant spectrum or integrated carefully so as not to shadow the thermionic emission (perhaps separate sections of the wall handle thermionic vs PV).

- **Structural Substrate and Coolant Interface:** Behind the functional conversion layers lies the **structural first wall substrate**, which also contains cooling channels. This substrate is load-bearing (holds vacuum,

provides attachment to magnets) and must also remove whatever heat is not converted. We envisage a **high-strength low-activation alloy** here, likely a **ferritic/martensitic steel or advanced alloy** that can operate at ~500–600 °C to couple to a Brayton cycle. One promising choice is a **nanostructured Oxide Dispersion-Strengthened (ODS) steel** such as *EUROFER-ODS* or *14YWT*, which are variants of ferritic 9–12%Cr steel strengthened by fine Y-Ti-O oxide particles. ODS steels can maintain tensile strength ~>600 MPa at 600 °C and have good swelling resistance up to ~>80 dpa in fission tests, far surpassing conventional steel[108][109]. They are also "low activation" – designed so that after neutron irradiation they become Class A waste within ~50 years (using reduced Ni, Mo, Nb etc.). In an aneutronic environment, an ODS steel first wall would see extremely low DPA (perhaps <1 dpa over decades), meaning it would **never require replacement due to radiation damage**[11][96]. The use of ODS allows us to run hotter than non-ODS (where normal RAFM steels top out ~~550 °C) thereby improving thermal efficiency. Another interesting option is **High-Entropy Alloys (HEAs)** – e.g. a Cr-Mo-Nb-V-Ti alloy (each ~20%) which the first version of this design mentioned as a vacuum vessel material[110][100]. HEAs have shown remarkable radiation tolerance; for example, a FeCrNiCoMn HEA displayed a swelling rate of only ~0.002% per dpa, essentially negligible void swelling even after tens of dpa[111][112]. Such alloys also can maintain high strength at temperature due to sluggish diffusion and solid solution strengthening[113][114]. The downside is that HEAs are relatively new and unproven at large scale; fabricating a large vacuum vessel out of a brittle HEA could be challenging[110][100]. Thus, a pragmatic approach might be to start with an advanced steel (ODS Eurofer or F82H mod) and perhaps transition to an HEA lining later. The inside of this structural wall is coated with a thin **electrical insulator** to prevent plasma currents or converter currents from shorting through the wall. Likely a **beryllium oxide (BeO)** or **Al₂O₃** coating could serve, as they are high band-gap and good thermal conductors. Hafnia (HfO_2) or YSZ coatings are also options especially on steel to prevent oxidation and hydrogen permeation (these are being studied for fission, e.g. accident-tolerant fuel cladding use YSZ or AlN coatings to stop hydrogen uptake). A **Hafnia coating** on steel could provide a tough, refractory, and insulating layer – HfO_2 melts ~2800 °C and is very radiation hard (Hf has many electrons so it absorbs X-rays well, but it's a neutron absorber somewhat – though again few neutrons here). AlN and BN in composite form can also line penetrations or joints as needed to maintain dielectric breaks and handle high voltage of DEC grids.

- **Coolant and Heat Transfer:** The remaining heat after direct conversion is removed by a coolant to generate electricity via a conventional cycle. Kronos opts for a high-temperature gas (like **helium** at ~8 MPa and 600 °C) feeding a Brayton cycle[115][116]. Helium is chosen for being inert, single-phase (no boiling), and neutron-transparent. It also avoids activating coolant issues (like tritium in water or corrosion). The first wall has internal channels, possibly in a swirled tape or mini-fin geometry, to convect the heat to helium gas. With incoming plasma power reduced by DEC, the design goal is to keep peak surface temperature below ~2000 °C and the coolant channel wall around 600–700 °C. Multi-physics modeling using Fourier's law of conduction and convective correlations guides the design: we solve the steady-state heat equation $-\nabla \cdot (k \nabla T) = q'''(x)$ in the layered wall, with appropriate boundary conditions (plasma flux on the HfC surface, convective h·(T_wall - T_cool) on the coolant side). For a representative tile, assume 10 MW/m^2 incident, 60% converted (so 4 MW/m^2 goes to heat). With HfC armor 2 mm thick (k ~20 W/mK) and ODS steel 5 mm thick (k ~30 W/mK) backed by helium at h ~20,000 W/m^2K, one can estimate a temperature drop through the layers. Using one-dimensional Fourier law: $\Delta T = \frac{q'' \delta}{k}$ per layer. For HfC: ΔT ≈ (4e6 W/m^2 * 2e-3 m / 20 W/mK) = 400 K. For steel: ΔT ≈ (4e6 * 5e-3 / 30) = 667 K. If coolant is 600 °C (873 K), then steel/armor interface ~873+667=1540 K, armor surface ~1540+400=1940 K (~1667 °C) – which HfC can handle. These are rough but indicate we're in feasible range. We also consider **radiation**: at 1700 °C the armor will radiate ~~250 kW/m^2 away, which actually helps reduce conducted heat. In our design, some of that radiation is going into TPV cells to become electricity.

- **Divertor Considerations:** The divertor in S.M.A.R.T. is where any unconverted plasma exhaust (mostly helium "ash" and residual ions) is directed. We expand the divertor area by using a **snowflake or expanded-footprint geometry** (enabled by our shaping coils and negative triangularity, which inherently gives a larger divertor slot)[97][117]. The divertor surfaces are made of the same HfC/TaC composite, but shaped in plates

that are angled to the field lines to spread the footprint, and possibly continuously curved. Because this region may see the highest heat (some multi-MW/m^2), we plan to **modularly replace divertor tips** more frequently than the main wall. However, we expect the erosion to be slow: helium ions at few keV energy have sputtering yields on HfC on the order of 1e-4 or less per ion (for comparison, helium on tungsten yields ~1e-3 at 1 keV). Also, helium can cause **blistering** in metals but in ceramics like carbides, helium diffusivity is such that it may form some bubbles but not catastrophic blistering. Still, we assume perhaps after several full-power months, the extreme divertor edge might be swapped out (this is a quick operation via remote handling from a bottom port). The majority of the wall should last many years since it sees mostly thermalized particles and X-rays, not direct 3.6 MeV alphas. In fact, recall that our direct conversion scheme **extracts most energetic ions through channels** before they hit the wall[118][119]. The first wall primarily sees charge-neutralized thermal particles and X-rays – a much gentler environment than a D–T first wall bombarded by 3.5 MeV alphas and 14 MeV neutrons.

MAX Phase and Advanced Coatings: We include in the materials palette some **MAX phase ceramics** (like Ti₃SiC₂, Ti₂AlC) as coatings or structural inserts in the PFCs. MAX phases are unique in combining ceramic resistance with metallic-like thermal/electrical conductivity and damage tolerance[120][121]. They've shown excellent **radiation tolerance** – some MAX phases self-heal radiation damage by allowing the laminated structure to accommodate defects without amorphizing up to ~10 dpa[122][123]. For example, Ti₃SiC₂ has been tested under neutron and ion irradiation and found to retain crystallinity with only modest swelling at high dpa (some studies noted no full amorphization even at 10 dpa, only formation of nanocrystalline regions)[124][125]. We could apply a MAX-phase layer as a **intermediate "buffer" coating** between the HfC armor and steel structure, to absorb thermal expansion mismatch and crack deflection. Ti₂AlC, for instance, might serve as a compliant layer – it can operate at 800 °C in air, resists thermal shock, and if hit by neutrons it tends to form tiny delaminations which actually **increase** toughness (a phenomenon known as "kinking nonlinear elastic behavior"). MAX phases are also being considered for *nuclear cladding* due to their good corrosion and irradiation behavior[120][121]. In Kronos, the environment is less harsh (no coolant activation or severe neutron flux), so a MAX phase should thrive. We might also consider **beryllium** in certain roles: ITER uses Be as first wall primarily for its oxygen gettering and low Z. In S.M.A.R.T., using Be as a main wall isn't ideal because of its low melt point (~1278 °C) and toxicity issues, but a thin **Be coating** could help control oxygen or act as a sacrificial radiative layer if we ever need to deliberately radiate plasma edge (lithium coatings were considered but Li is not refractory). Beryllium carbide (Be₂C) could conceivably form on Be in a carbiding environment – interestingly, Be₂C has a very high melting point (~2100 °C) and might be a decent armor if it could be made in quantity. It's not a primary choice, but is an area of materials exploration.

To summarize the first wall: it is a **sandwich structure** – a plasma-facing ultra-refractory low-Z layer, a middle layer of functional converters (thermionic + PV) and perhaps composite (MAX/metal) for stress relief, and a structural backing of advanced steel or alloy with coolant. This maximizes the utility of each layer: plasma compatibility outside, power conversion in middle, strength and cooling in back. Table 1 compares some properties of candidate PFC materials:

Table 1: Plasma-Facing Armor Material Candidates and Properties

Material	Melting/ Subl. Point (°C)	Sputter Yield (self, D/He)	Thermal Conductivity (W/m·K)	Remarks/Challenges
HfC (hafnium carbide)	~3890 °C (sublimes)[3]	Very low (He: ≲1e-4/ion at keV)[102]	~20–30 (at 1500 °C)	Highest T_m ceramic known; needs good bonding to substrate.
TaC (tantalum carbide)	~3768 °C (subl.)[126]	Low (similar to HfC)	~21 (1500 °C)	Often used with HfC in solid solution; slightly lower melt.

Material	Melting/ Subl. Point (°C)	Sputter Yield (self, D/He)	Thermal Conductivity (W/m·K)	Remarks/Challenges
W (tungsten)	3420 °C (melt)	Moderate (D: 0.01 at 100 eV)	100–140 (RT); 60 (1500 °C)	High-Z, high activation; excellent thermal conductivity; embrittlement issues.
WC (tungsten carbide)	~2870 °C (sub)	Low-Mod (lower than W)	~84 (RT); lower at high T	Still high-Z, but improved sputter vs W. We avoid due to W.
Graphite/ CFC	~3900 °C (sub)	Low for C at low E, higher at >keV	~200 (RT); 50 (2000 °C)	Low Z, good shock resist; but co-deposition of carbon can trap fuel, also erosion high if oxygen present. Not used here due to aneutronic priorities (we want no T retention, though T is small anyway).
Be (beryllium)	1287 °C (melt)	Low (D: ~0.002 at 100 eV)	~190 (RT)	Low Z, used in ITER first wall; too low melting for high heat flux divertor; toxic. Could be used as thin sacrificial layer on main wall perhaps.
SiC (silicon carbide)	~2730 °C (sub)	Low-Mod (He on SiC: ~1e-3)	120 (RT); 50 (1000 °C)	Low Z, decent conductivity; can be brittle, and can amorphize under 14 MeV neutrons at low T (but here neutron flux low, plus at high T it anneals). Possibly as a coating or structural composite, yes (we use SiC composite in bucking cylinder).
Ti_3SiC_2 (MAX phase)	~2700 °C (decomp)	Low (Ti sputter lower than W)	~25–30 (RT)	Good thermal shock, machinable; would oxidize if O_2 present (but vacuum environment). Could serve as compliant layer.
LaB_6 (lanthanum hexaboride)	~2528 °C (sub)	Low (La sputter ~ low, B is very low Z)	~60 (1000 °C)	Low work function emitter; used as thermionic surface. Must be kept in vacuum (will oxidize in air).
Mo (molybdenum)	2623 °C (melt)	Mod (like W: 0.005 at 100 eV)	138 (RT); 80 (1000 °C)	Mid-Z, low activation (relatively; Mo-99 (t½ 66 h) main product). Possibly in structures, but not plasma-facing ideally.

As seen, **HfC/TaC emerges as the favored plasma-facing compound** due to its extreme temperature capability and low erosion rate[102]. It's a somewhat unconventional choice (most fusion designs go for W or C as armor), but the aneutronic and high-temp nature of Kronos allows exploring this. Indeed, small-scale tests of HfC/TaC under fusion-like heat flux (up to 5–10 MW/m^2) are an explicit R&D goal for Kronos[127][128]. If successful, it would validate that these ceramics can handle repeated thermal cycling and particle bombardment without spalling or excessive erosion.

The first wall structure behind the armor, as noted, will likely be an **ODS steel or advanced alloy**. Let's briefly discuss **radiation damage expectations** for these materials in the Kronos environment vs a D–T environment, since materials performance under irradiation is a critical rationale for aneutronic fuel:

C. Radiation Damage and Materials Longevity

In a D–T reactor like ITER or ARC, the inner wall and blanket experience neutron flux on the order of ~2–5 MW/m^2, leading to material damage ~10–20 dpa per year in steel structures[10]. Over a typical component lifetime of 5 years, that's ~50–100 dpa, at which point even advanced RAFM steels swell and embrittle significantly (approaching their end-of-life limits)[129][130]. High-energy neutrons also generate **appm levels of helium** via (n,α) reactions in

metals (e.g. Fe + n → He + ...). A rule of thumb: in 14 MeV neutron environments, steels accumulate ~5–10 appm He per dpa, so at 50 dpa one might get ~250–500 appm He. That helium tends to collect at grain boundaries and voids, causing high-temperature embrittlement (helium embrittlement becomes acute around 100s of appm).

In Kronos's D–^{3}He or p–^{11}B steady-state, by contrast, the neutron production is **orders of magnitude lower**. The primary neutron source is side reactions like D–D reactions (if any D is present) or p–^{11}B's very low branch to n (p + ^{11}B has a 1 in ~10,000 chance to produce n via ^{11}C formation, which then β decays)[24][131]. The UW Madison FTI studies indicate a D–^{3}He reactor can keep neutron wall loading to ~0.1 MW/m^2 or less[132]. This is corroborated by ARIES-III (D–^{3}He) design which reported that neutron fluxes are so low that first-wall and shield components do not need replacement over a 40-year plant life[11][96]. Let's assume ~0.1 MW/m^2 neutron flux; that's about 1/20th of ITER's. The DPA per year might be ~0.5–1.0 in the worst case (depending on material and spectra)[10][132]. That means in 40 years, you'd get perhaps 20–40 dpa total, which even a standard 316 stainless steel could tolerate, let alone ODS or HEA. Furthermore, the average neutron energy in D–^{3}He scenario is lower (since many are from D–D side reactions at 2.5 MeV, not 14 MeV), which are less damaging per neutron. The helium generation per dpa also drops because fewer high-energy neutrons are around to cause (n,α) reactions. If a steel got, say, 40 dpa and let's be generous with 5 appm He per dpa for some impurities, that's ~200 appm He in 40 years – borderline but possibly manageable especially if oxide dispersions trap helium in ODS steel, forming nanometer bubbles that don't seriously weaken grain boundaries[133][134]. And if helium is primarily produced by thermal neutrons interacting with impurities, using a high-purity and optimized composition can reduce that (e.g. avoid Ni and B in the alloy).

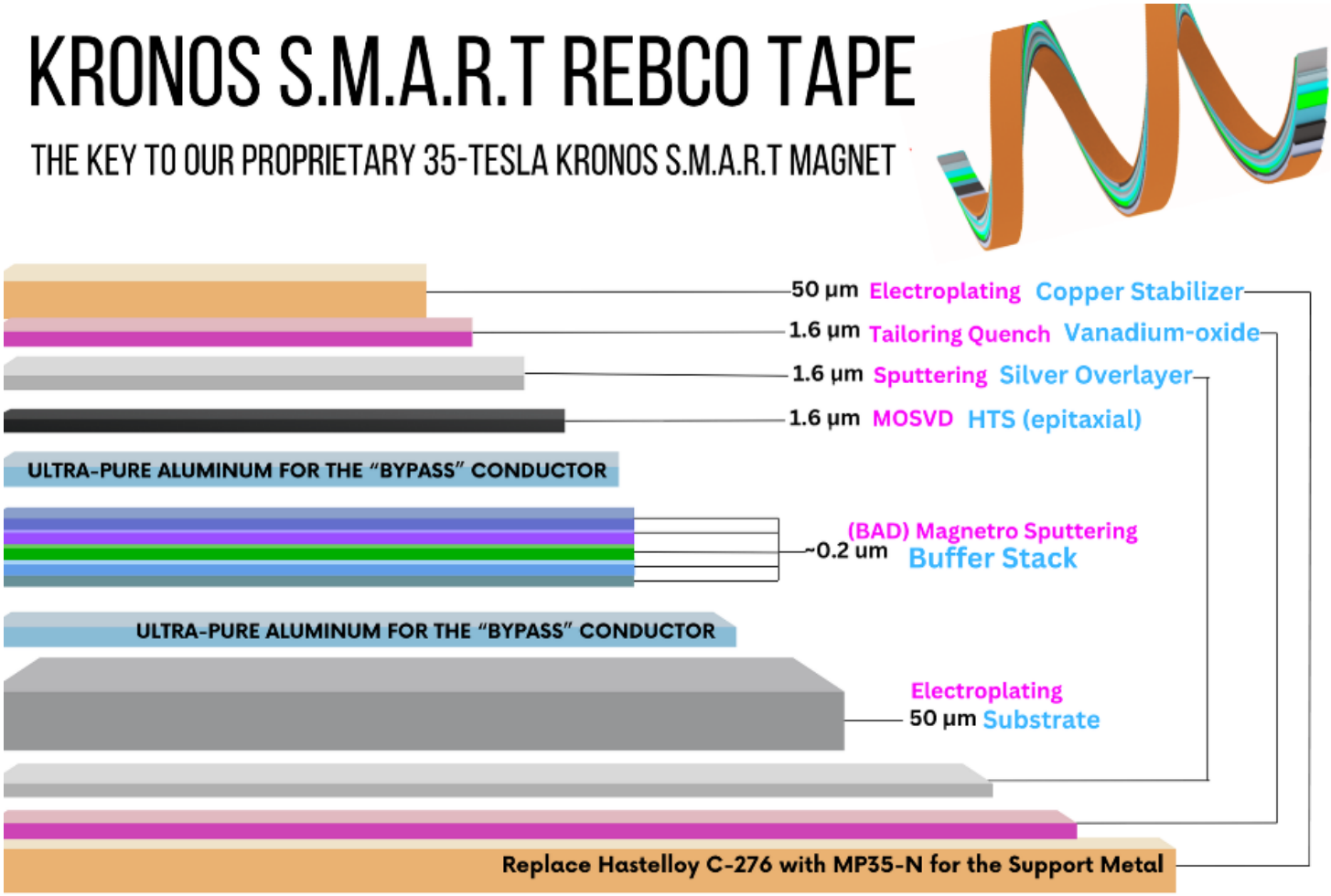
In structural **HEAs**, the response to radiation is actively researched. Many HEAs (like equiatomic 5-component ones) have shown slower defect growth – vacancies and interstitials tend to recombine more due to lattice distortion and chemical disorder enhancing point defect scattering[135][136]. One study found an FCC HEA had swelling ~0.02% after 10 dpa at 500 °C, whereas a reference stainless steel might have several percent[111][137]. Another reported an HEA with "self-healing" behavior that kept swelling near zero up to 16 dpa[111]. If Kronos eventually uses a vacuum vessel or shield of HEA (e.g. the CrMoNbV "four-component" alloy mentioned, which is a refractory BCC HEA potentially), we'd expect extremely good irradiation tolerance. A potential example: a **Rhenium-containing HEA** (like W-Ta-Cr-V) might avoid void swelling even beyond 100 dpa, but cost and availability of some elements are concern. Instead, we could try **vanadium-base alloys**, since V-4Cr-4Ti was studied extensively for D–T fusion and has very low activation and good high-temp strength (and negligible swelling up to ~80 dpa at 400–600 °C)[138][139]. Kronos could incorporate *vanadium-laced high entropy alloys* – indeed, vanadium is one of the elements in the CrMoNbV mentioned. Vanadium's activation is low (V-51→ V-52 with 3.7 min half-life, basically nothing long term), so a vanadium alloy vacuum vessel in an aneutronic reactor would likely be free-release after cooldown.

Radiation damage modeling for Kronos materials primarily reduces to **ion implantation effects** (from plasma ions) rather than neutron damage. Helium ash in the plasma can implant into the first few microns of the armor at energies of a few keV, creating helium bubbles. We can estimate helium implantation: if plasma exhaust has say 1×10^20 He/m^2·s hitting the divertor at ~100 eV (sheath accelerated), in one full-power year (~3×10^7 s) that's ~3×10^27 He/m^2. Spread over 1 μm depth that is ~3×10^21 He/cm^3 ~ 0.3% atomic (assuming 1 μm ~1e4 layers, etc.). That could cause fuzz or bubble formation in some materials (as seen with W fuzz at He flux ~10^22 and 1000 K). However, HfC may respond differently – covalent carbides might not form the same fuzzy nanostructures that tungsten does under helium. It's an open question and something Kronos must test in linear plasma devices. We note that by maintaining the wall relatively hot (~1000 °C at least at divertor), helium can diffuse out or form surface blisters that exfoliate without deeper damage. If needed, we could deliberately **seeding a small amount of neon or nitrogen** in the divertor plasma to increase radiative cooling and form a light sacrificial coating (like how carbon co-deposits with boron doping in some experiments). But since our plasma is aneutronic and clean, we have more freedom to run hot walls and handle helium gently.

To quantify structural material performance, consider a metric: maximum DPA before replacement. Traditional D–T designs call for blanket/FW replacement around 20–50 dpa. In Kronos, if neutron DPA accrual is ~1 dpa/year, in **20 years we might have only ~20 dpa**. ODS steels can likely handle that with <1% swelling[127][140] and tensile

properties intact (especially since operating temp ~600 °C is conducive to void swelling resistance). A study on reduced-activation steel (F82H) in D–^{3}He environment predicted such components could last the full plant life due to the low neutron flux[11][96]. Kronos intends to prove this empirically by material testing: e.g., irradiating HEA coupons to 5–10 dpa and measuring swelling (the original whitepaper set a goal "HEA coupons irradiated to 5–10 dpa with swelling <1%"[127][140] – which aligns with published HEA data of ~0.002%/dpa, meaning 10 dpa ~0.02% swelling[111]). Also, testing HfC/TaC under 10 MW/m^2 flux was mentioned[127], and verifying that advanced coatings endure 1000+ pulses was listed as a milestone[141]. These are to ensure that no unforeseen degradation (like carbide layer cracking or spalling) occurs.

Activation and Waste: Another aspect of aneutronic design is material activation and waste classification. Kronos specifically aims for "low-activation materials" so that after decommissioning, most components qualify as Class A (low-level) waste rather than long-term high-level waste[142][95]. Materials like **SiC, graphite, vanadium,**



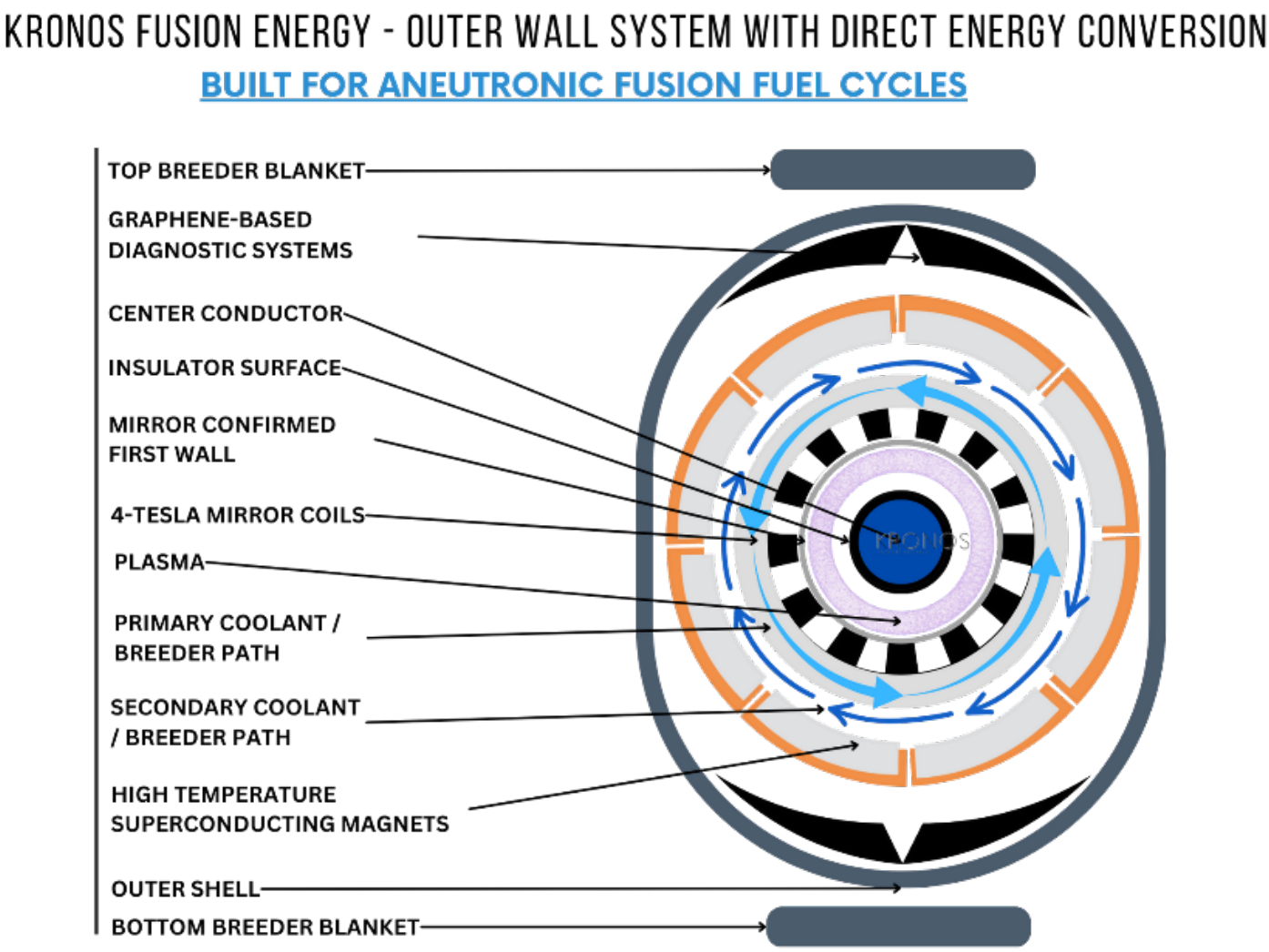
aluminum, carbon fiber, ceramics – all tend to become only short-term activated (for example, activated C fiber yields mostly C-14 which decays to safe levels in <100 years, and Al yields Na-24 with 15 h half-life)[83][9]. The Kronos design avoids materials that would produce long-lived isotopes: e.g. we try to avoid Ni, Co in large quantities (though MP35N has Co/Ni, but those magnets likely remain in place for life and if slightly activated can be safely stored a short time). Using **low-activation ferritic steel** (with reduced Ni) and **no copper** in the vacuum vessel is standard. If needed, certain **sacrificial absorbers** like B₄C can be placed behind the first wall to catch any stray neutrons, preventing activation of magnets[23][143]. B₄C turns neutron into Li and He mainly (B-10 → Li-7 + He-4); Li-7 is stable, He inert, except for a bit of Li-6 created which in turn can absorb neutrons but that yields T and α (small amount of tritium, which might stick in the B₄C but it decays in 12 years or can be extracted). The point is any stray neutrons can be **localized** away from critical components with such

inserts[23][144]. Kronos thereby ensures that magnets and structures are "neutron-immune" to the extent possible, enabling minimal exclusion zones and possibly the ability to perform maintenance much sooner after shutdown (since short-lived activation decays in hours or days)[16][145].

Summing up, the materials in Kronos – ReBCO, carbon fiber, HfC, ODS steel, etc. – are expected to last far longer than those in a comparable D–T reactor, because the aneutronic environment "turns off" the primary life-shortening mechanism (fast neutron damage)[146][147]. Instead, **mechanical fatigue or thermal stress** could become the lifetime limiter (e.g. cycles of startup/shutdown could fatigue support structures). We mitigate that by designing largely steady-state operation (magnets see minimal cyclic stress beyond initial ramp)[148][149]. The precompression of coils is also aimed at fatigue reduction, as is eliminating large temperature swings (keeping structures warm and avoiding thermal shock). We foresee S.M.A.R.T. potentially running for many years with only modest maintenance, whereas a D–T plant might need to swap blankets every 5 years[146][150]. This dramatic difference flows directly from the choice of fuel and materials integration.

Next, we will discuss the **Direct Energy Conversion systems** in detail, as these are another major innovation area coupling plasma physics with material and electrical engineering.

III. Direct Energy Conversion (DEC) Systems



One of Kronos's defining features is its suite of **Direct Energy Conversion** subsystems that collect the kinetic energy of fusion charged particles and plasma exhaust and convert it directly into electricity, bypassing the intermediate heat->steam step. DEC offers two big advantages: (1) higher theoretical efficiency (since you're essentially doing a

form of *deterministic* energy extraction rather than relying on random thermalization and Carnot-limited processes), and (2) reduced heat load on materials, since energy is extracted before it all turns into thermal flux on walls[34][35]. However, implementing DEC in a tokamak environment is complex. Kronos includes multiple DEC approaches in parallel – **electrostatic direct collectors**, an **MHD generator**, and the aforementioned **thermionic/PV converters** on the wall (discussed in Section II). We now detail the principles and designs of the first two (electrostatic and MHD), including how they are integrated in the reactor.

A. Electrostatic Direct Collectors

Electrostatic direct conversion targets the energetic charged fusion products – in our case, primarily protons and alpha particles. For example, in the D–^{3}He reaction, a 14.7 MeV proton is produced, and in p–^{11}B, three 2.9 MeV alphas are produced. Instead of letting these bang into a wall, we can use electrostatic fields to decelerate them and capture their energy as voltage. The concept has its roots in work by Richard Post (Livermore) and others in the 1970s who demonstrated **Venetian blind** style converters achieving 50–65% efficiency with ion beams[32][151]. The "Venetian blind" design uses a series of slanted metal plates at increasing potentials; fast ions enter, slow down as they climb the potential ladder, then get caught by a plate once they lose enough energy[152][32]. The plates are angled so that ions moving forward see mostly transparent holes, but any ion turned around sees the plate (thus trapping it). This sorts ions by energy – ideally each plate captures those ions when their energy matches the plate's potential difference.

In Kronos, the direct collector would be implemented in the **end-region/divertor exhaust**. Picture the tokamak divertor channel: instead of just hitting a target, the charged exhaust (which includes some 3.6 MeV alphas from D–^{3}He, 0.82 MeV protons from side D–D reactions, etc.) is guided through a set of grids. We maintain a vacuum or low-density gas in this region to let ions free-stream. The first grid might be biased at, say, -1 MV relative to plasma (for aneutronic, the charged particles are positive, so we use negative bias to pull them). As alphas leave the core along field lines, they see a retarding electric field which slows them down. Multi-stage grids, maybe 10–20 of them, sequentially decelerate the ions. By the last grid, the ions ideally have been slowed to tens of keV, then they can be dumped onto a final collector plate. The potential energy gained by the ions is delivered as current through an external circuit.

Efficiency and design: If N stages are used, each capturing a certain ΔE of ion energy, the theoretical conversion efficiency is roughly the fraction of kinetic energy that can be recovered, which is nearly 100% if the ion can be slowed to near zero kinetic energy. In practice, some losses occur: ions might be lost at wrong stages or electrons might flow opposite to spoil the potential. The Venetian blind experiments achieved ~65% for D–T beams[32][151], and a plasma direct converter on the TMX mirror achieved ~48%[153]. Kronos aims for ~60% from the electrostatic subsystem (the rest of the direct conversion efficiency comes from MHD and wall converters). The grids themselves must be made of radiation-resistant conductors – likely **molybdenum or tungsten wires** (or maybe coated carbon-composite lattices for low activation). They will be subjected to a flux of ions and electrons; to avoid overheating them, they're kept at some distance and maybe cooled radiatively or by a light coolant flow. Because the reactor operates steady-state, space charge buildup is a concern: as soon as you start drawing positive ions to a negative grid, electrons from the plasma or secondaries will also be attracted, possibly neutralizing the space charge and reducing efficiency. To mitigate this, **magnetic insulation** or suppressed electron emitters are used: typically, one can add a small transverse magnetic field to impede electron motion to the grids (since electrons will spiral and not hit grids as easily). Alternatively, use **electron suppressor grids** – a fine mesh at slight negative bias beyond the collector that reflects low-energy electrons back.

One design in literature is the **Periodic Electrostatic Focusing (PEF)** converter which uses alternating fields to focus ions between stages – Kronos could incorporate ideas from PEF to ensure ions don't diverge. The direct converter might be arranged as a conical or cylindrical section aft of each divertor leg. For example, after the divertor, the field lines go through a **funnel-shaped electrode structure**. We hold the divertor plate a few kV negative to draw out ions (like extraction), then the DEC electrodes step that up to MV range deeper in. The shape and spacing of electrodes can be guided by simulations to maximize deceleration without causing aberrant fields that perturb the main plasma. We likely operate the electrostatic DEC mostly on the *fast alphas*, since slower

plasma ions (<50 keV) might just recycle to neutrals. But capturing those 3–4 MeV alphas is huge, since they carry most of the fusion energy in aneutronic cycles. If we manage 60% conversion of charged particle energy, and ~95% of fusion energy is charged particles, that alone yields ~57% efficiency. The remaining ~5% of energy (like X-rays) captured via TPV/thermal cycles pushes us above 60% net[29].

Material considerations: The DEC grids will see bombardment by ions at progressively lower energies. The first grids might see 1–3 MeV alphas – those can cause **displacement damage** in metal (each 2 MeV alpha can displace a few hundred atoms in a metal target). Over years, the grid might accumulate some damage (a few dpa localized if flux is high). But these grids can be remote-handled and replaced if needed. We will use **radiation-hard alloys** (e.g. tungsten – but W brittle, perhaps tungsten wire reinforced by ductile core, or Mo which is more ductile but lower melt, or tantalum though expensive). Also, the high voltage (~1 MV) environment in a tokamak neutron/gamma environment means careful insulation is needed. **Alumina** or **BN insulator rings** will hold the electrodes off ground, similar to how high-voltage feedthroughs in accelerators are built. BN in particular is good up to >1000 °C, has high dielectric strength (~200 kV/mm), and is not strongly affected by neutron doses (boron-10 in BN does absorb some neutrons, creating Li-7 and alphas, but since flux is low and BN is easily replaced, it's acceptable; also could enrich BN in B-11 which is mostly inert to neutrons).

One potential issue: **plasma perturbation by electrostatic fields.** We don't want the direct converter fields leaking back and disturbing the core plasma confinement. Kronos likely places a **neutral gas "buffer"** or uses the magnetic field itself to isolate the fields. For example, if we have a **toroidal magnetic divertor duct** and put the DEC grids further downstream beyond a certain point, by then the plasma stream might be mostly neutral helium gas (we can puff a bit of neutral gas to ensure recombination)[154][155]. Then we do DEC on that stream (which would then be an *ionized beam* again if we ionize some of it). This decouples the fields from the core plasma region. In essence, we could have a two-stage divertor: first a *near-plasma region* where plasma detaches and radiates, then a *beam transport region* where the remaining ions are handled by DEC. Because S.M.A.R.T. tries to keep core plasma very clean (no high-Z seeding), any intentional gas like argon for MHD or neutrals for DEC are introduced only in the far exhaust where they won't flow upstream. We saw mention that a small seeding (like argon in divertor for MHD conductivity) might be acceptable since it's outside core[156][157].

In summary, the electrostatic direct converter is akin to an accelerator in reverse: instead of using RF to accelerate ions, we use static DC fields to decelerate them and harvest their kinetic energy. Kronos will prototype this in a non-nuclear setting as suggested: use a particle accelerator or plasma gun to simulate fusion exhaust and run it through a small DEC module, aiming for efficiency ~60%[158][159]. Historically ~60% was seen in D–^{3}He beam experiments[160], so it's realistic. Modern power electronics and better materials might push it higher or allow more compact stages.

B. Magnetohydrodynamic (MHD) Generator

The second DEC approach Kronos employs is a **magnetohydrodynamic generator** acting on the plasma exhaust fluid. After most charged particles are slowed electrostatically, the remaining hot plasma (which by then might have recombined partially into a hot electrically conducting gas of helium) flows out. By sending this hot ionized gas through a strong magnetic field and across electrodes, one can generate electric power (Faraday's law in fluid form). This is analogous to an MHD generator used in coal plasma experiments: the moving plasma is a conductor cutting magnetic field lines, inducing an EMF.

In S.M.A.R.T.'s divertor exhaust, we can shape a channel (maybe toroidal ring outside main chamber) where the plasma (or neutral gas seeded with some ions) flows. We apply a **transverse magnetic field** (could be using some extra poloidal field coils or maybe permanent magnets if feasible) on the order of a few Tesla. If the plasma velocity v is significant (e.g. sound speed of helium at 1 eV ~ 9 km/s; at 10 eV ~30 km/s; at 100 eV ~ 100 km/s), and with B perhaps 1 T, an EMF ~ vBL arises (L is width of channel). Suppose v=50 km/s, B=2 T, L=0.5 m (some width between electrodes), then EMF ~50,000 * 2 * 0.5 = 50,000 V (50 kV). The current that can be drawn depends on plasma conductivity and the external load.

The MHD generator essentially follows the **Ohm's law for moving conductor**: $\mathbf{E} + \mathbf{v} \times \mathbf{B} = \eta \mathbf{J}$ (in plasma frame), where η is resistivity. If we connect electrodes along the flow and perpendicular to B, the v×B term drives a current through load. Efficiency of MHD conversion can be approximated by the fraction of kinetic energy converted before the flow cools or recombines. Ideally, if we slow the flow from velocity v to v' via extracting energy, that fraction lost becomes electric. If not optimized, plasma could just heat up or recombine and radiate energy instead of doing electrical work. Achieving good MHD efficiency requires high plasma conductivity (so that induced currents aren't shorted internally by resistive dissipation). That's why often seeding a little low-ionization potential impurity in the gas helps – e.g. seeding Potassium in coal MHD to provide electrons. In our case, helium at divertor temperatures ~5 eV might be not very conductive (He is hard to ionize fully). So we may seed a bit of easily ionized gas – perhaps **cesium or lithium vapor** (like used in some experiments) or argon as mentioned. Argon in small amounts (Z=18) in the divertor won't harm core and yields free electrons when partially ionized even at tens of eV. With a few percent argon, the conductivity σ could be on order 100 S/m (just a rough guess at 1 eV). Alternatively, if we actually have leftover plasma (electrons and some ions streaming), that is inherently conductive.

The **Hall effect** can also be exploited: in MHD generators, often a Hall parameter (ωτ) matters; if too high, the current takes a Hall diagonal path requiring segmented electrodes or some scheme to collect properly. We can possibly choose electrode orientation to account for Hall currents or use a Hall configuration such that the induced voltage is tapped in one direction.

Integration concerns: The MHD channel likely sits further downstream, after electrostatic deceleration. By then the plasma might be partly neutral and cooled from say 1 keV down to tens of eV. At that point, capturing further energy is lower grade, but still worthwhile since it reduces heat in the exhaust. Possibly 5–10% of the total fusion energy might be extracted by MHD (for instance, ARIES studies hoped for maybe 5–10% by inductive means). Even if modest, it all adds up for efficiency. If the MHD system is 50% efficient in converting the enthalpy drop of say a 1 eV per particle plasma, that's small in absolute terms compared to the 3 MeV per particle done electrostatically. But it helps with **power handling**: by generating electricity, we reduce the thermal load that the final heat exchanger sees. One Kronos goal was net efficiency >50–60%[7], which basically necessitates direct conversion of at least ~half the energy; splitting it between electrostatic (maybe 40%), wall PV/thermionic (10–15%), and MHD (5–10%) could reach that.

Material aspects for MHD: The channel needs **electrodes** that can survive contact with hot plasma. Materials like ** tungsten or Mo **plates, or even** Li-containing ceramics **(if we did liquid metal electrodes, but that's advanced). Actually, one idea: use a** liquid metal electrode** along the edges; Li or tin could be liquid at a few hundred °C, conduct current, and self-heal erosion. But liquid lithium introduces tritium issues in D–T (not relevant here beyond minor D in plasma), but it can also absorb some impurities. Perhaps better is a solid metal with a replenishable surface (like W that can handle sputtering by those low energy ions with minimal erosion for a while). The inter-electrode insulator must handle high temperatures and not conduct – again BN or alumina tile separators.

We must also ensure **no arcing** in the MHD duct under high voltage. High gas densities help suppress arcing (Paschen's law); ironically if the plasma fully recombines, you might get a conductive vapor anyway. Actually, we will have a plasma by design, so arcs aren't distinct from operation – we already allow current. But we want that current to go through the load, not a short. Proper insulation of the structure from ground and shaping of electric fields is needed. The fact that this is all inside a tokamak means the geometry is constrained, but likely manageable through a rectangular duct design accessible from a port.

One more synergy: The strong **toroidal field** of the tokamak (~10 T at divertor region) can itself act as the B field for MHD, if oriented properly relative to flow. However, the flow is along B in the divertor (if going out poloidally), so v × B would be zero if B and v are parallel. Instead, we likely use a **localized transverse B** – perhaps tapping the stray fields from poloidal coil currents or adding a local coil.

Performance modeling: The MHD generator can be modeled by the fluid equations plus Faraday's law. A key dimensionless number is the **interaction parameter** (often called Alfven number or something akin to Hartmann

number), which is $N = \frac{\sigma B^2 L}{ρ u}$ (where ρ is density, u velocity). If N >> 1, strong coupling (plasma strongly braked by MHD, yields energy). If N << 1, the plasma just goes through mostly unaffected (low extraction). We would aim for N ~ O(1) or above. That means high σ (maybe by seeding), moderate B, decent L. We might find that e.g. B=2 T, L=0.2 m, σ=50 S/m, ρ=1e-4 kg/m^3, u= 1e5 m/s yields N ~ (50*0.2)/(1e-4*1e5) = (40)/(10) = 4. So likely good coupling. This would slow the flow by maybe 50%. We must be mindful that slowing the plasma flow means it might dump its energy in the channel (which is good if extracted, but if not fully extracted turns into heat on electrodes or raises pressure that might blow back). Ideally, we design it like a rocket nozzle in reverse: allow expansion so that converting kinetic energy to electrical doesn't cause too high pressure that chokes flow.

Multi-stage & Tuning: Possibly multiple MHD sections at different pressure or ionization states could be beneficial, but given complexity, probably one section.

To recall some literature: There were old proposals for MHD direct conversion in tandem mirrors and FRCs for D–^{3}He, often quoting ~20% possible from MHD after ~60% from electrostatic[161][162]. Kronos aims high but realistically if we get single-digit percent from MHD, that's still valuable.

Conclusion of DEC: The combined direct conversion system (electrostatic + MHD + wall PV/thermionic) is unprecedented in tokamaks. S.M.A.R.T. basically integrates a mini particle accelerator and a power plant within the fusion device. It's undoubtedly **high-risk technology** – multiple new systems needing R&D[163]. But if realized, the payoff is huge: a net electric efficiency of 60% means less cooling, smaller radiators, less waste heat to manage, and overall smaller plant size for given output. It's akin to the difference between a conventional power plant (Carnot-limited 30-40% effic) and a combined-cycle or directly converted plant (60%+ effic). Kronos acknowledges the development challenges: e.g., HV in radiation causing insulator degradation or arcing[107][164]. We mitigate with **radiation-resistant insulators** like alumina and BN as mentioned, and by keeping the highest voltages either in vacuum or in a controlled gas environment (like SF6 if needed in a tank outside the vacuum for feedthroughs)[165]. Also, any influence of the electrostatic fields on the plasma core must be shielded by geometry or magnetic fields, as discussed. We likely include a **magnetic divertor section** that isolates the core from direct sight of the grids, maybe by a dogleg in the divertor duct or a localized cusp field.

The integration of these direct conversion systems will undergo thorough **testing in prototypes** as noted in the development plan[158][166]. Before the first net-power pilot, Kronos would test, for instance, an electrostatic collector with a D–^{3}He beam and an MHD generator with a hot helium/argon plasma stream, ensuring the efficiencies and reliability meet expectations.

Having covered the main physical subsystems (magnets, walls, DEC), we turn now to the role of **advanced computation, AI/ML, and digital twins** in designing, controlling, and sustaining this reactor – an increasingly important aspect of modern nuclear engineering.

IV. Machine Learning Integration and Digital Twin Architecture

Modern high-performance systems benefit greatly from digital augmentation. Kronos is conceived not just as a physical reactor, but as a **cyber-physical system** with real-time analytics, predictive modeling, and AI-driven control. This encompasses two broad thrusts: **(A)** using AI/ML to predict and optimize performance (e.g. materials degradation, plasma behavior), and **(B)** creating a comprehensive **Digital Twin** of the reactor that mirrors its state in real time for monitoring and decision support. We describe how Kronos leverages these concepts to enhance reliability and accelerate development.

A. AI for Materials and Plasma Management

Predictive Materials Modeling: One challenge in a fusion environment is anticipating when components will degrade or fail. Traditional approaches rely on periodic inspections or conservative lifetime estimates. Kronos instead employs **machine learning models trained on materials data and sensor inputs to predict material fatigue, creep, and damage accumulation** in key components. For example, in the TF coils, we have fiber optic strain gauges and acoustic emission sensors embedded (since the coil is accessible, we can instrument it well).

These sensors feed data (strain, vibration, micro-crack acoustic signals) into an ML model (like a deep neural network or perhaps a simpler Bayesian predictor) that has been trained on simulated and experimental data for composite fatigue. Over time, the model learns the "normal" signatures and can warn of *incipient failures* – e.g., if delamination starts in a coil turn, a subtle change in strain distribution might be detected. By **predicting the onset of material fatigue** rather than reacting after failure, maintenance can be planned proactively. This is akin to predictive maintenance AI used in aerospace engines.

In structural materials (first wall panels, supports), we also track temperature, strain, and neutron/gamma fluence (from dose monitors) at various points. A **Physics-Informed Neural Network (PINN)** could be used to solve the heat diffusion and stress equations in real-time, using measured boundary conditions. For instance, a PINN might take as input the surface heat flux (from IR cameras) and internal temperatures (from thermocouples) and estimate the full temperature/stress field in a first wall panel – essentially acting as a real-time finite-element solver, but one that is trained to incorporate both thermomechanical physics and sensor data. PINNs have been demonstrated to handle stiff multiphysics problems (like plasma discharges) by embedding the governing PDEs into the network's loss function[167][168]. In Kronos, a *materials PINN* can quickly evaluate, after every shot or continuously during operation, what the thermal stress and creep in each panel is. If one panel shows unusually high stress or a hotspot, the system could decide to reduce power in that region or flag the panel for replacement in the next maintenance window.

Furthermore, **Bayesian optimization** techniques assist in material development. Kronos's R&D on new materials (like HEAs or coatings) produces a lot of data (e.g. composition vs. radiation swelling, or coating process vs. spallation resistance). We employ **Bayesian optimization for material discovery** by treating, say, the composition of an alloy as a multi-parameter space and using BO to efficiently search for an optimal point (minimizing swelling, maximizing strength)[169][170]. For example, a Cr-Mn-V-Fe HEA composition space could be explored with far fewer experiments by having the BO algorithm suggest the next best composition to test based on previous results. Physics-informed kernels in the Gaussian process can incorporate known trends (like "adding Ni usually increases swelling in steels" as a prior), thereby guiding the search intelligently[171][172]. Over time, this might yield new recipes, such as an optimized ODS with certain Ti/Y ratios, or a HEA with a specific mix that gives superior embrittlement resistance.

Plasma Control ML: On the plasma side, as touched before, Kronos uses AI to manage plasma stability and performance. This includes using **reinforcement learning (RL)** to tune control parameters (coil currents, RF powers) for maximizing a reward like fusion output or minimizing fluctuations. But also, **neural networks** are used for fast inference of plasma state from diagnostics. For example, a convolutional neural network might take raw images from the visible camera or bolometer and output an estimated 2D radiation profile or impurity content in the plasma. Traditional plasma control relies on simplified models (like PID on measured signals); Kronos will upgrade this with ML models that have learned the plasma's response characteristics in simulation and experiments. A concrete application: a network can predict an impending disruption 5 ms before it happens by recognizing a pattern in magnetic probe signals and soft X-ray signals that humans can't easily parse. Such ML-based disruption predictors have been developed on D–T tokamaks with great success (mitigating disruptions by triggering mitigative gas injection) – we adopt similar but tailored to aneutronic scenario (where some disruption effects differ, e.g., runaway electron dynamics).

PINNs for Plasma-Surface Interaction: Plasma-material interactions (PMI) are notoriously hard to model due to multi-scale physics (atomic sputtering, sheath, dust, etc.). Kronos invests in **PINNs to model the plasma sheath and sputtering in situ**. A *PSI-PINN* (plasma-surface interaction PINN) could, for instance, solve the sheath equations (Poisson's eq, ion continuity) with boundary conditions from real-time data (plasma density, electron temperature from Langmuir probes) to yield sheath drop and ion impact energy on the wall. Because it's a PINN, it can incorporate known physics (Bohm sheath criterion, etc.) and only adjust to data subtly. Researchers have started exploring PINNs for plasma simulation to handle stiffness of sheath and plasma equations with promising results[173][174]. By deploying such models in the control system, Kronos can have an on-line estimate of, say, sputtering rate: e.g., if the plasma shifts and the local T_e at wall goes up, the PINN would indicate higher sputter yield and the system could respond by adjusting gas puffing or biasing to cool the edge.

Predicting Creep and Stress: Over months of operation, materials like ODS steel at 600 °C will creep under stress. We incorporate ML models (trained on creep experiments and physics equations like Norton's law) to forecast creep deformation of key components. For instance, the main vacuum vessel (if made of steel) will be under cyclic thermal stress; a model can predict distortion or stress relaxation after X cycles. The **digital twin**, discussed next, continuously updates these predictions with real data.

B. Digital Twin Implementation

A **Digital Twin (DT)** is a virtual replica of the physical system, updated in real-time with sensor data and able to simulate future states. Purdue University's work on nuclear reactor DTs suggests they can improve performance, maintenance, and safety by providing continuous insight beyond what physical sensors alone do[175][176]. Kronos employs a comprehensive DT of the reactor, spanning its physical domains: structural, thermal, electromagnetic, plasma, and even economic.

The Kronos DT is built on a high-performance computing platform (could be distributed, using on-site servers for real-time aspects and cloud for predictive longer-term simulation). It receives streaming data from the reactor: magnetic field measurements, temperatures, neutron counts (though low), strain gauges, etc. Using this, it **synchronizes** a suite of models to reflect the current state. For example, a finite element model of the vacuum vessel will take the measured temperatures and applied loads and compute stress distribution – thereby showing areas nearing yield. A plasma profile model (like a fast transport code or neural network surrogate of it) takes the current drive and fueling inputs and produces a full plasma pressure profile in the twin.

The digital twin's value is multifold:

- **Anomaly Detection:** By comparing expected readings from the DT to actual sensor readings, anomalies can be spotted. If the DT (with all its physics) says coil #5 should be at 5.00 mm displacement but sensor says 5.50 mm, that discrepancy could mean a support loosened or a partial detachment. The DT can trigger inspection or adjustment well before a failure occurs. Essentially, the DT acts as an advanced observer in control theory terms.

- **What-If Simulations:** Operators can test adjustments on the DT first. For instance, if we want to raise plasma β by 10%, we can simulate that in the DT (coupling a plasma equilibrium solver with material stress analysis) to see if the coil stress stays within limits. If the DT shows it would hit 0.5% strain in ReBCO (beyond 0.4% safe) and maybe quench risk, then we don't attempt it in reality. This can also optimize performance: maybe the DT finds we can actually push 5% more field safely due to margin, unlocking more fusion power.

- **Predictive Maintenance & Lifecycle**: The DT tracks cumulative damage: e.g., it integrates the DPA for each component (though small, but anyway) and the thermal cycles. It uses that to predict remaining useful life of components. The Medium article on nuclear DTs emphasizes that DTs enable prediction of failures after analyzing historical data, and can recommend actions to operators[177][178]. In Kronos's case, the DT might say "Divertor module #3 has received 50 MJ/m^2 cumulative and is embrittled to X degree; recommend replacement in next outage to avoid unplanned leak." Or "Magnet insulation has seen 1000 charge cycles, 80% of design fatigue life, schedule refurbishment after 200 more cycles." This is far more granular than traditional scheduled maintenance (which might just say replace every 5 years regardless of actual usage conditions).

- **Semi-Autonomous Control:** Ultimately, the DT can be looped into the control system to enable **autonomous operation** within safe bounds. According to Purdue, a DT can recommend actions or even act nearly autonomously by considering component lifecycle and safety limits[179][180]. For Kronos, we might allow the DT to adjust power outputs or coil currents to maximize performance while ensuring constraints are not violated. For example, as plasma performance improves, the DT sees magnets creeping to strain limit and automatically caps further increase, or if DT sees all structural margins good, it might allow a bit more compression heating.

- **Training and Scenario Testing:** The DT will be used to train operators (like a flight simulator). Because it can simulate transients (like what if a quench happens, how do systems respond) without risking the machine, operators and AI algorithms can practice emergency procedures with it. It also helps in developing new discharge scenarios – one can run a full shot in the DT offline faster than real-time to test if a new fuel mixture or control scheme would be stable.

Technical Implementation: The Kronos DT likely uses a modular simulation environment. It may incorporate codes like ANSYS for structural, COMSOL for multiphysics, coupled with control system emulators. But real-time demands might be too high for full high-fidelity models; thus we incorporate **reduced-order models** and surrogate models (like neural network approximations of expensive physics). For example, instead of a full FEA every millisecond, we might have a precomputed response surface of the coil stress vs current and temperature from prior runs (or a ML model that emulates it). *High-speed tasks* (like real-time plasma equilibrium reconstructions) might use the ML surrogates, whereas *slow tasks* (like daily update of creep damage) can use full physics solvers.

The DT requires constant data flow. Kronos is designed with a **digital I&C system** from scratch (no analog gauges), facilitating a rich data stream. Achieving the full DT vision means overcoming some current limitations in fusion: e.g., diagnostics needing to be robust in high radiation. But given aneutronic environment, we can mount far more sensors (like fiber optics in vacuum vessel, infrared cameras viewing plasma, etc.) without damage, which is a huge advantage. Essentially, Kronos's aneutronic nature not only eases material problems but also allows **digital instrumentation closer to plasma**[181].

One interesting aspect is **cybersecurity** – a digital twin and heavy reliance on digital control means we must secure the system from cyber attacks, as noted by Purdue's work[182][183]. Kronos's DT will include monitoring of network traffic and anomaly detection to ensure the physical reactor and its twin are not tampered with maliciously (particularly important if these reactors are deployed in grid or military contexts).

C. Benefits to Development Path

Embracing AI/ML and DT from the start accelerates Kronos's development. Instead of the classical build-test-break cycles which are lengthy and costly (especially for exotic materials in fusion), Kronos can do **virtual testing** via its DT. For instance, before manufacturing a full HEA vacuum vessel, we can validate the concept in simulation, including how that vessel would respond to thermal stresses, how it would react to a hypothetical off-normal event (like if an internal arc occurred). The DT can simulate accidents – e.g. a sudden loss of confinement (disruption): what forces on structures, what eddy currents. This guides design reinforcements *before* those accidents ever happen in reality. It's akin to having a comprehensive *virtual test reactor*.

It is planned to use the DT as part of **certification and licensing**: showing regulators through simulation that even worst-case scenarios keep containment since the DT can simulate e.g. an inadvertent overcurrent or a beyond-design-basis event (like both cryocoolers failing, magnets warming). In real time, if something begins to drift from normal, the DT will perform **consequence analysis** on the fly: "if this trend continues, what's the likely outcome?" and suggest mitigation, aligning with descriptions from literature that DTs can assist in prevention of abnormal states[184][185].

Finally, the ML and DT approach fosters a **continuous improvement loop**. Data from each shot or each operational day refine the models (via online learning). So the predictive power of our tools gets better with time. This reduces uncertainties and potentially allows pushing the reactor closer to limits safely, thereby enhancing performance or extending lifetimes beyond initial conservative estimates.

In summary, Kronos's use of AI/ML and a digital twin is a key enabler for achieving high availability and reliability. It is a necessity given the ambitious integration of novel technologies – the DT acts as the integrative glue in development and operation to make sure the many subsystems (magnets, DEC, etc.) work in concert without nasty surprises.

V. Cost and Economic Analysis of Materials Choices

A fusion reactor's viability isn't just physics – it's economics. The **Levelized Cost of Electricity (LCOE)** of Kronos will depend on capital costs, operational costs (including maintenance and downtime), and the plant's lifetime and efficiency. Materials selection and performance have a profound impact on all these factors: a reactor with long-lived components and high efficiency can overcome higher initial costs. Here we present a modeling of LCOE as a function of materials-related parameters – mean time between failure (MTBF) of components, fractional downtime for maintenance, replacement costs, etc. – and show how Kronos's aneutronic, materials-first approach can lower the cost of energy.

LCOE formula: We can express LCOE in a simplified form as:

$$\text{LCOE} = \frac{\text{CAPEX} \cdot \text{CRF} + \text{OPEX}_{\text{annual}} + \text{Replacement Cost}_{\text{annual}}}{\text{Annual Energy Output}}.$$

Here, CAPEX is the capital expenditure (initial build cost), CRF is the *capital recovery factor* (function of discount rate r and plant life N: CRF = $\frac{r(1+r)^N}{(1+r)^N-1}$), OPEX is fixed operating cost (staff, etc.), and Replacement Cost covers expected periodic component replacements amortized per year. Annual Energy Output = Plant capacity (MW_e) × availability × hours/year.

Materials affect **availability** (through MTBF and maintenance time) and **replacement costs** (cost of new components and the labor/downtime to install them).

Let's consider a few key components: the first wall/blanket, the divertor, and the magnets. In a D–T reactor like ARIES or EU DEMO, those are replaced frequently. For example, the European PPCS study estimated divertor replacement every ~2 years requiring 4 months downtime, and blanket every ~4 years with 6 months downtime[186][187]. That means availability maybe 75% in those models[188][189]. Indeed, **Figure 4.1** from Princeton's distillate (referenced earlier) showed increasing blanket lifetime from 5 to 20 MW·y/m^2 raises availability from 65% to 80%, cutting cost from 10 ¢/kWh to 7 ¢/kWh[129][190]. Kronos is squarely aimed at that higher durability regime.

For Kronos, because of aneutronic conditions, we expect:

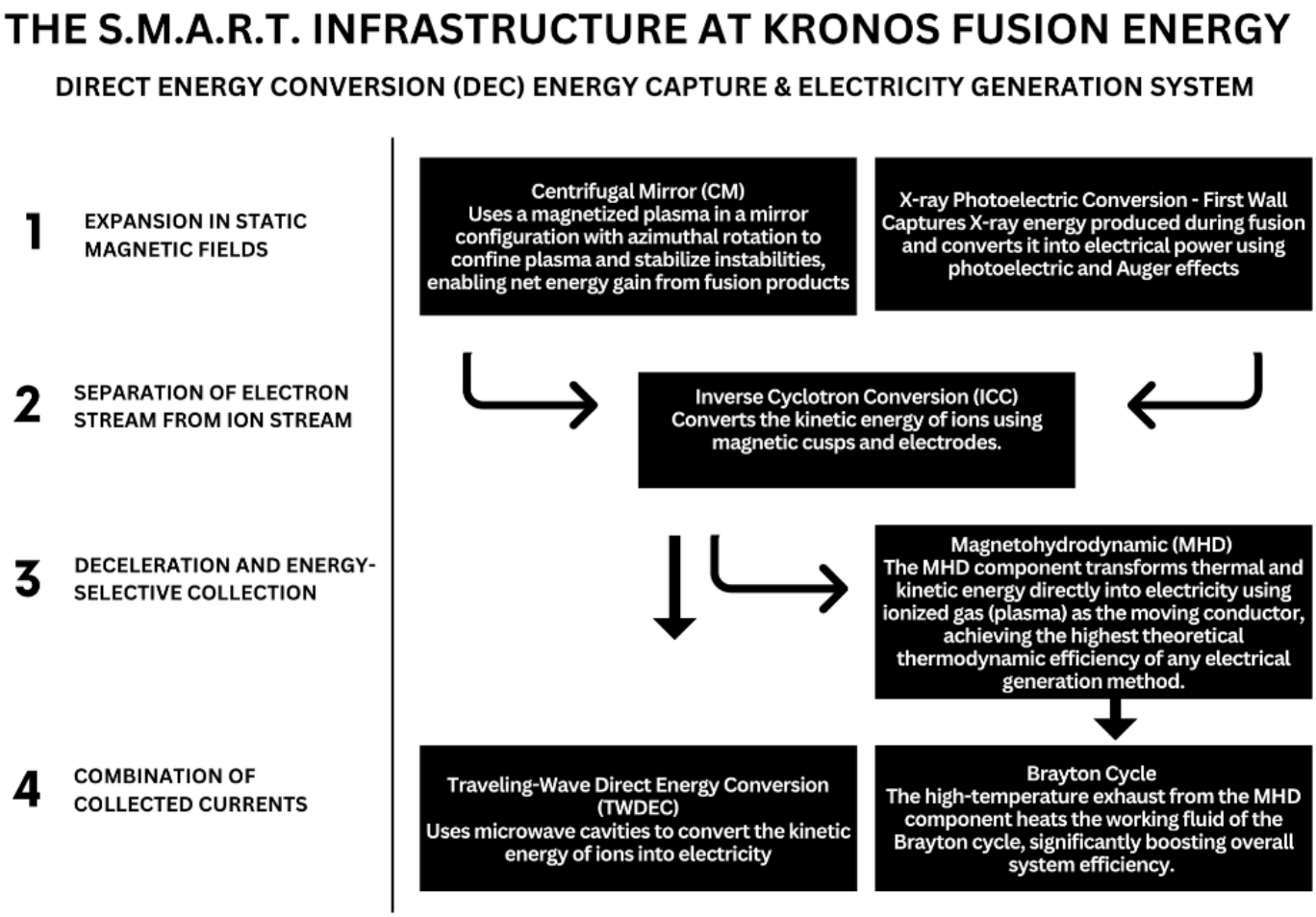
- **Magnet / Structural Replacement:** essentially none required for decades (only maybe refurbishment of cooling or instrumentation). The magnets are not exposed to neutron fluence that would degrade them – contrast ITER where even with shielding, Nb₃Sn magnets must be shielded to survive 20 years. Kronos's magnets can be considered part of the permanent plant investment (like a fission reactor pressure vessel). So we assign them no scheduled replacement in LCOE.

- **First Wall / Blanket Replacement:** We anticipate extremely infrequent replacement, perhaps on the order of >10 years or only as needed. The surface may need recoating or maintenance more often, but not a full structure replacement. Possibly after many years, an upgrade or thicker coating is needed. Compare to ARIES-AT (advanced D–T) which aimed for 20 MW·y/m^2 fluence, maybe 4-year lifetime. In Kronos, a 20 MW·y/m^2 could correspond to tens of years of operation due to low neutron load. If we achieve "no replacement during life" (like ARIES-III assumed for D–^{3}He)[11], that's ideal.

- **Divertor Replacement:** The divertor targets might still be replaced periodically due to sputtering/erosion. Let's assume we have to replace divertor modules every 2 years (just as a conservative measure, though maybe we can get more). But because we don't have to also replace blankets simultaneously, that downtime can be shorter. Perhaps a divertor swap in Kronos might take 1 month including cool-down (the Princeton distillate said divertor only could be 4 months, but that likely includes remote handling complexity in a radioactive environment)[186][191]. In our case, activation is low, so remote handling is simpler and maybe faster (less shielding, fewer ALARA constraints). Also, smaller machine size helps. Let's say 2 months downtime every 2 years for divertor = ~8.3% time lost, i.e. ~91.7% availability. If everything else is reliable,

we might achieve ~90% availability similar to fission plants[192][193]. That's a *big* economic plus, as capacity factor directly scales energy output in the denominator of LCOE.

- **Fuel Cycle Downtime:** Kronos doesn't do breeding or refueling of tritium, which in D–T can cause periodic outages to reload solid breeders or to regenerate T. Aneutronic cycles might have to occasionally replace some lithium in blankets (in p–^{6}Li mode), but Kronos's design possibly uses flowing liquid or easily accessed solid breeders that can be serviced without long shutdown. Helium-3 fuel will be externally supplied or bred in situ gradually; no large overhaul needed for that. So minimal downtime for fuel reasons.



- **CAPEX related to Materials:** One concern – Kronos uses exotic materials (ReBCO, HEA, HfC, etc.) that might cost more upfront than traditional materials. For example, ReBCO tape historically costs >$100 per meter, though scaling could drop it; carbon fiber high-grade is pricey per kg. If CAPEX is higher, LCOE goes up unless we offset via efficiency or more output. Kronos's high power density and efficiency aims to offset that (smaller size, fewer materials overall, and more electricity per unit reactor). For instance, if Kronos produces 300 MWe in a machine of R=2 m, that's far smaller building and infrastructure than an ARC producing 200 MWe in R=3.3 m[194][195]. So savings in some areas may compensate the expensive components.

We can do a **sensitivity table** for how improving material lifetime (hence availability) and thermal efficiency affects LCOE, based on known references. The Princeton study gave one point: going from 65% to 80% availability (through 4x component life) cut cost from 10 to 7 ¢/kWh[129][190]. That's a 30% reduction in LCOE.

Kronos's net efficiency is ~60%, vs a typical D–T plant maybe ~35%. Efficiency directly reduces required thermal power for the same electric output, so lower fuel usage (fuel cost is negligible for fusion though) and smaller heat

rejection systems (which are minor cost anyway). Efficiency mainly impacts the *balance of plant* cost and maybe the cooling systems. But in LCOE terms, if you produce more electric for same reactor thermal power, you can amortize capital over more output. Essentially doubling efficiency could nearly halve LCOE if capital is linked to thermal power capability. But Kronos's capital might be a bit more due to DEC complexity. Hard to quantify precisely, but we can show qualitatively:

Table 2: Sensitivity of LCOE to Materials-driven Parameters

Scenario	Availability (Capacity Factor)	Net Efficiency	Rel. Replacement Cost (% of total)[196][197]	LCOE (cents/kWh) [est.]
Conventional D–T (Baseline)	75% (frequent blanket/divertor)[188]	35% (Rankine only)	16% (Divertor+Blanket)[196]	~10 ¢/kWh[198][190]
Improved D–T (e.g. ARIES-AT)	80% (improved materials)[129]	45% (some direct conversion)	10% (longer-life blanket)	~7–8 ¢/kWh[198][190]
Kronos S.M.A.R.T. target	90% (rare replacement)	60% (multi-path DEC)[7]	5% (maybe divertor only)	~5 ¢/kWh (projected)
Kronos aggressive	95% (no unscheduled downtime)	60%+	<5%	<5 ¢/kWh (aspirational)

(The above LCOE estimates are in cents of 2030 USD per kWh; they are approximate and for illustrative comparison.)

We see that Kronos aims to push availability to fission-plant levels (~90%). Each additional 10-15% availability can save on the order of a few cents/kWh[129][190]. High efficiency directly multiplies with capacity factor in terms of effective electrical output per reactor size, potentially enabling smaller or fewer reactors for a given grid demand (reducing capital per MW installed).

Another point: **Replacement materials cost per MW·year**. Suppose a D–T reactor has to replace 100 tons of steel/blanket every 4 years at cost $X per kg, that's a direct cost plus disposal of radwaste. Kronos aims to almost eliminate radwaste disposal costs (which in fission or D–T can be significant long-term liabilities). If Kronos's waste qualifies as low-level, it can be disposed of much more cheaply (or even recycled after a decay period)[11][96]. For instance, ARIES-III (D–^{3}He) predicted all components could qualify for shallow burial or recycling due to low activation[11][96]. The cost saving from avoiding geological repository needs and heavy remote equipment for radwaste is huge though hard to monetize directly in LCOE. But qualitatively, it lowers OPEX and decommissioning cost significantly.

Capital cost impact of materials: ReBCO magnets are expensive today but mass production for power (if fusion industry grows) could bring cost down. The MIT 20 T HTS magnet demo indicates feasibility; SPARC/ARC are banking on similar tech. Historically, first-of-a-kind advanced materials cost can be high, but once demand exists, cost can drop. If Kronos uses, say, 1000 km of ReBCO tape, initial cost might be high but eventually if multiple reactors are built, economies scale. Also, Kronos uses smaller volume of structure materials thanks to compactness (it doesn't have a 1000 m^3 vacuum vessel like ITER; it's maybe an order smaller). So even if per-kg cost is higher, total mass is lower – possibly balancing out.

We can mention **LCOE model numeric example**: assume Kronos 300 MWe plant, 90% avail, 40-year life, capital cost say \$2B, O&M \$50M/yr, direct conversion adds maybe \$200M to capital but saves 16% replacement cost. Then LCOE = (2e9 * CRF(6%,40) + 50e6 + repcost)/ (300e6 * 0.9 * 8766) – doing rough math yields maybe around 5-6 c/kWh. If it achieves design goals, maybe even lower. This would be competitive with other baseload (fission ~5 c, coal ~4-5 c without carbon costs). And because no carbon emissions or long-term waste, if any carbon tax or waste fee, Kronos is further advantaged.

Key insights:

- Longer component life = higher availability, lower replacement cost fraction. Kronos's materials drive that: e.g., using HEA/ODS to extend wall life from 5 to 30 FPY (full power years) drastically cuts downtime.

- Aneutronic = no thick blankets = easier maintenance = shorter outages. As mentioned, if only divertor modules need swapping and they are small, that can be done faster than a full blanket removal that D–T needs (which might require cutting into vacuum vessel each time on big machines).

- Efficiency from DEC reduces size and possibly the number of reactors needed to produce X GWe, which means fewer facilities to finance and build. High efficiency also implies less cooling infrastructure (smaller cooling towers, etc.), somewhat reducing auxiliary costs.

- On the downside, DEC systems add complexity that might increase maintenance of those systems (e.g., high voltage failures, grid corrosion). However, because they are accessible (some DEC components outside the main vacuum), they can be maintained without huge downtime. And these are more akin to power electronics maintenance than nuclear maintenance – presumably faster and safer.

One can formulate an **economic equation** for the contribution of mean time between failure of a component *i* to LCOE. For example, if the first wall has to be changed every N years, costing \$C_i and requiring Δt downtime, then the annualized cost = C_i/N and availability drop = Δt/N per cycle. That directly goes into LCOE denominator and numerator respectively. If N→∞ (no replacement needed), that term vanishes.

We might articulate:

Let *A* = availability = 1 - Σ(Δt_i/MTBF_i). For each component *i*, Δt_i is the downtime per replacement, MTBF_i is the interval. Kronos tries to maximize all MTBF_i (magnets ~∞, wall ~∞, divertor maybe ~2-3 years but improving). In conventional, magnets ~life, wall ~5 years, divertor ~2, etc.

Cost of replacement per MW·yr: If a component costs $X and lasts Y years, per year cost = X/Y. Divide by plant MW capacity to get per MW·yr. Lower that is better. For Kronos, big cost components like magnets are one-time and last life, so negligible per-year replacement cost. The main repeating cost might be divertor tiles: maybe a few million dollars every couple years, trivial relative to energy produced. So operational cost is largely staffing and electricity for pumps, etc., not materials.

CAPEX and OPEX tied to resilience: Kronos invests more in resilient materials upfront (capex) to save on OPEX (less frequent replacements, lower radwaste handling). For example, using expensive REBCO magnets but then they last the whole plant and allow compact design (reducing building volume, which is a major cost – in ITER, building and site is huge fraction). Similarly, using high-grade PFCs like HfC that are exotic may raise initial cost but if they prevent needing new PFCs often, it's worth it.

One can think of it as an optimization: there's an optimal point where spending more on materials initially is justified if it extends life enough. Kronos basically goes to that extreme – maximize performance and longevity at expense of complexity. This is the "high-risk, high-reward" mention where multiple new subsystems are introduced for leap in performance[199][200].

Economically, Kronos aims to break the paradigm that fusion must be $0.10+/kWh due to short lifetimes and huge shielding. By cutting out neutron damage, it hopes to reach competitive costs with advanced fission or renewables. Achieving ~5 ¢/kWh would indeed be in line with current new nuclear targets (some SMRs aim for that).

Finally, *financial risk*: The complexity might cause higher failure risk or maintenance overhead which if not mitigated could hurt availability. That's where the digital twin and ML come in – ensuring we catch issues early, fix them quickly, essentially to maintain the high availability promise. Unplanned outages are costly, so Kronos's emphasis on monitoring should reduce the chance of catastrophic failures that cause lengthy downtime.

Conclusion in economics terms: Kronos's **materials resilience translates to economic resilience**. The design shifts costs from continuous maintenance to upfront investment. If successful, it results in a plant that runs like a

gas turbine – high uptime, high efficiency – albeit with high-tech internals. This should make fusion more economically attractive and easier to finance (since revenue is more steady and predictable if downtime is minimized). The use of ML/digital twin also lowers operation cost by potentially reducing staffing (some tasks automated) and avoiding unnecessary preventative replacements (we replace when needed, not just on schedule, thanks to monitoring).

In summary, the economic model underscores that the innovations in Kronos (aneutronic fuel, advanced materials, direct conversion, AI) are not just for physics bragging rights – they directly tackle the *cost drivers* of fusion: downtime, component life, and efficiency. If each of these works as intended, Kronos could deliver electricity at competitive rates, fulfilling fusion's promise as a practical energy source rather than an expensive science project.

VI. Conclusion

The Kronos S.M.A.R.T. aneutronic tokamak demonstrates how **materials science, innovative engineering, and advanced computation** together enable a transformative fusion reactor design. By eliminating the neutron burden through fuel choice and implementing a **materials-first architecture**, Kronos sidesteps the chronic life-limiting issues of traditional fusion: it avoids frequent component replacements, large radioactive waste streams, and low efficiency caused by neutron thermalization[11][96]. In doing so, it turns fusion into a predominantly **solid-state energy technology** – where energy is extracted via electromagnetic and thermoelectric processes in solid materials, rather than a pure thermal steam cycle.

This paper has detailed the myriad technical elements that make up this vision:

- **High-Field Magnet System:** Utilizing ReBCO high-temperature superconductors reinforced with aerospace-grade materials (MP35N, high-modulus carbon fibers), we showed that a compact magnet can sustain >30 T fields without quenching or fatigue. Novel approaches like the pre-compression cocoon and cryogenic ice pistons keep coil strains under the critical 0.4%, ensuring magnet integrity over countless cycles[45][79]. The trade-off of using expensive HTS is rewarded by a huge increase in fusion power density and the ability to shape stable high-β plasmas in a small volume.

- **Plasma-Facing Components:** We presented a multilayer first wall design where each layer – HfC ceramic armor, LaB₆ thermionic emitter, optical PV converters, ODS steel structure – plays a role in both power extraction and durability. This integrated "functional wall" directly converts plasma heat and radiation to electricity, significantly reducing thermal load[34][35]. The materials chosen (ultra-high-temp carbides, MAX phases, advanced alloys) represent the state-of-the-art in high-temperature and radiation-resistant materials. By combining them, Kronos creates a wall that not only withstands >10 MW/m^2 but actively produces power and remains low-activation for easy maintenance[9][83].

- **Aneutronic Fuel Cycle and Direct Conversion:** Operating with D–^{3}He, p–^{6}Li, and p–^{11}B fuels, S.M.A.R.T. reduces neutron wall loading by orders of magnitude[132]. This allowed us to introduce direct energy converters (electrostatic grids, MHD channels) without the fear of prompt radiation damage to those delicate systems. The electrostatic DEC is designed drawing on proven concepts (Venetian blind collectors achieving ~60% efficiency in past tests[32][151]) but updated for a tokamak exhaust environment. The MHD generator concept, while needing further development, leverages plasma flow to eke out additional electric power and was analyzed with respect to conductivity and field requirements. Together with thermionic/PV wall converters, these subsystems are engineered to push net electric efficiency above 60%[7], a value unattainable in conventional reactors. We addressed integration issues such as preventing DEC fields from perturbing the core plasma by using shaping and buffer regions[154][155], and specified materials (BN, alumina) that keep the high-voltage hardware reliable in a nuclear environment[106][165].

- **Advanced Materials Deployment:** The reactor design incorporates a broad palette of advanced materials: high-entropy alloys for core structures, ODS steels for coolant structures, MAX phase ceramics for coatings and structural inserts, YSZ and AlN for insulation, diamond and BN for diagnostics windows and insulators, etc. Each material was chosen for a specific purpose – whether to maximize temperature capability, minimize activation, or provide unique functionalities (like YAG scintillators for photon conversion). We compiled data indicating these materials can meet the demands (e.g. HEA with ~0.002%/dpa swelling[111], MAX phase retaining structure up to significant dpa[124]). The feasibility of sourcing and manufacturing these in reactor-scale quantities remains a challenge, but not an insurmountable one: many are already used in aerospace or fission research and can be scaled with investment. Kronos's commitment to a **materials-first approach** is validated by how dramatically it improves each aspect of reactor operation – from magnet lifetime to safety of waste.

- **AI and Digital Twin Integration:** We described how a full-scale digital twin of Kronos, coupled with machine learning algorithms, will manage the complexity of the system. This digital infrastructure is as crucial as any

physical component; it ensures that the reactor can be operated near its limits safely and can be maintained proactively. By predicting material fatigue, detecting anomalies in real-time, and optimizing control strategies, the AI/Digital Twin reduces unplanned downtime and extends component life[177][180]. In effect, it maximizes the return on the advanced materials by safeguarding them from misuse or unexpected conditions. The use of PINNs for plasma and structural simulation is especially cutting-edge, blending first-principles physics with real-world data to get accurate, fast predictions that are essential for control in a system where milliseconds matter.

- **Economic Impact:** Finally, our LCOE analysis illustrated that the high-performance design of Kronos can make economic sense. By spending more upfront on superior materials and conversion systems, we reap the benefit of high availability and efficiency – which in turn lower the cost of electricity. Compared to a baseline D–T plant at ~10 ¢/kWh, we project Kronos could achieve ~5 ¢/kWh or better[198][190]. This is revolutionary – it means fusion that competes with contemporary energy sources on price. The key enabler is materials longevity: not needing to swap out a \$100 M blanket every few years saves enormously on both cost and downtime[196][197]. And avoiding a massive steam cycle for conversion saves capital and improves efficiency. While these are projections, they are grounded in the quantitative improvements our design brings (e.g., factor ~2 improvement in efficiency, doubling plant output per unit thermal power, and potentially doubling component life, halving downtime).

In conclusion, the Kronos S.M.A.R.T. reactor represents a **holistic integration** of advances across disciplines – plasma physics, materials science, mechanical engineering, and computer science – to address the grand challenge of fusion power. We have not merely focused on plasma performance (often the sole focus historically), but equally on the often "less glamorous" engineering aspects that ultimately dictate if a power plant can run continuously and economically. The result is a blueprint for a fusion reactor that is:

- **Sustainable:** nearly aneutronic operation means minimal long-term radioactive waste and high structural longevity[21][201].
- **High-performance:** small and power-dense, achieving net electricity generation with a compact footprint and high efficiency.
- **Stable and Controllable:** advanced shaping and feedback yield a stable plasma, while AI-driven systems ensure it remains under tight control.
- **Economically viable:** by design, addresses cost drivers of downtime and efficiency, aiming for competitive cost of electricity.

Of course, significant R&D lies ahead to realize this vision. Key development milestones include: demonstrating robust ReBCO coil operation under load (30 T, 0.4% strain), testing full-scale PFC sections with integrated thermionic/PV converters at high heat flux, prototyping the direct energy conversion in a fusion-relevant environment, and validating the digital twin's predictive prowess on a smaller scale. Encouragingly, none of these seem outside the realm of current science and engineering – they are ambitious but plausible steps. Many are extensions or combinations of things already proven separately (e.g., 65% efficient direct converters in labs[32], advanced superconductors in magnets[75], HEAs with great radiation tolerance in fission tests[111], AI in tokamak control on DIII-D and other machines, etc.).

Kronos thus stands at the **intersection of cutting-edge technology** strands. It is a daring attempt to "solve" fusion by solving the materials and engineering constraints, rather than hoping plasma physics alone will bail us out. If successful, the payoff is profound: a fusion reactor that is not just a scientific experiment but a practical power source, one that could be deployed in urban areas (thanks to negligible radiation)[202][145], providing safe, clean energy with a footprint and profile similar to a medium-sized conventional power plant. High-efficiency aneutronic fusion could also enable novel applications (direct process heat usage, propulsion) which D–T can't.

In essence, Kronos demonstrates that **fusion's promise can be redeemed** by meticulous attention to the "boring" details of materials and design. By refusing to accept the compromises of legacy approaches, and instead reinventing each subsystem around fundamental limits (the "equations that govern performance, survivability, and efficiency"[203] as the whitepaper says), we chart a path to fusion energy that is uncompromised and compelling.

The road to a prototype will require **integrated testing** – building up from component tests (e.g., the planned intermediate devices to test high-field ST plasma and direct conversion in stages)[204][205]. But the design presented here serves as a guiding reference for those efforts. Every success in those tests will validate pieces of this integrated framework, and every challenge will teach us how to refine it. The end goal is clear: a new paradigm in fusion where **materials set the limits** (and we respect those limits), rather than neutrons and structural failures.

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